

FUNDAMENTALS OF THERMODYNAMICS & HEAT TRANSFER

PROPERTIES OF COMMON SUBSTANCES



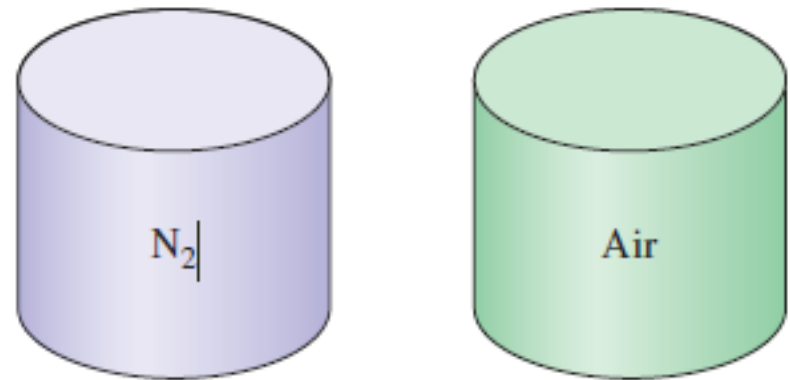
Er. Sanjeev Karki
Deputy Head & Assistant Professor
Department of Applied Sciences
Advanced College of Engineering & Management

Learning Objectives

- Introduce the concept of a pure substance.
- Explain the state postulate.
- Describe the hypothetical substance “ideal gas” and the ideal gas equation of state.
- Explain and illustrate two phase system in the $P - v$ and $T - v$ property diagrams.
- Introduction with the different terms used in two phase system.
- Understand the properties of two phase mixture.
- Introduce concept of quality and moisture content.
- Demonstrate the procedures for determining thermodynamic properties of pure substances from tables of property data.

Pure Substance

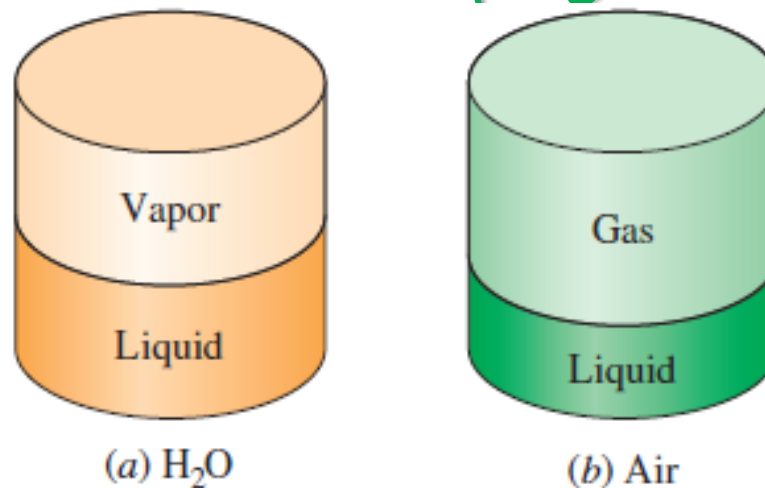
- A substance that has a fixed chemical composition throughout is called a **pure substance**. Water, nitrogen, helium, and carbon dioxide, for example, are all pure substances.
- However, pure substance does not have to be of a single chemical element or compound.
- A mixture of various chemical elements or compounds also qualifies as a pure substance as long as the mixture is homogeneous.
- Air, for example, is a mixture of several gases, but it is often considered to be a pure substance because it has a uniform chemical composition.



Nitrogen and gaseous air are pure substances.

More on Pure Substance

- A mixture of two or more phases of a pure substance is still a pure substance as long as the chemical composition of all phases is the same.
- A mixture of ice and liquid water, for example, is a pure substance because both phases have the same chemical composition.



A mixture of liquid and gaseous water is a pure substance, but a mixture of liquid and gaseous air is not.

More on Pure Substance

- A mixture of liquid air and gaseous air, however, is not a pure substance since the composition of liquid air is different from the composition of gaseous air, and thus the mixture is no longer chemically homogeneous.
- This is due to different components in air condensing at different temperatures at a specified pressure.

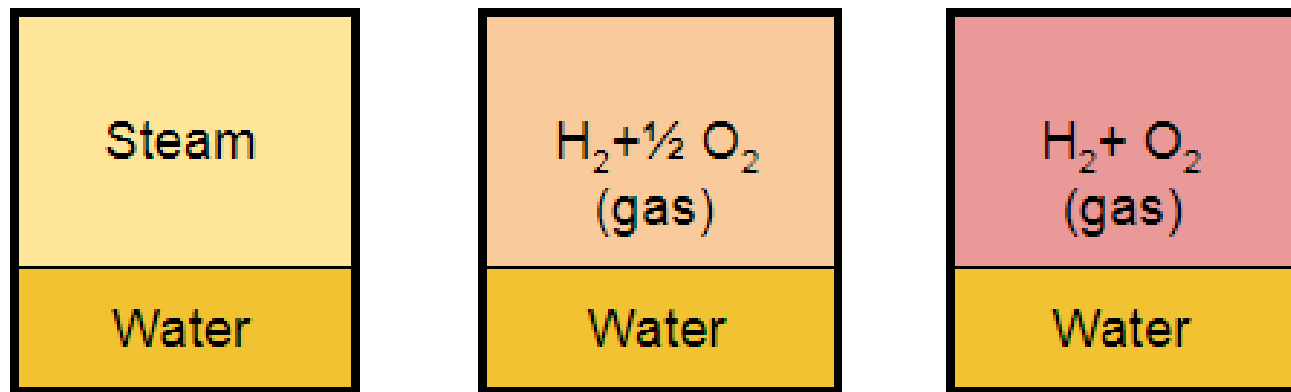
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More on Pure Substance

- For most of the thermodynamic analysis, we assume the working substance as a pure substance.
- A system is said to be composed of **pure substance** if it is
 - a) homogeneous in composition
 - b) homogeneous in chemical aggregation, and
 - c) invariable in chemical aggregation
- **Homogeneous in composition** means that the proportion of the constituent elements of the substance should be same throughout the system.
- **Homogeneous in chemical aggregation** means that the constituent elements of the substance should be combined in the similar manner in all samples of the system.
- **Invariable in chemical aggregation** means that the state of homogeneity should not change with time.

Illustration of definition of pure substance

- Let us consider three systems, system 1 consisting of homogeneous mixture of liquid water and water vapor, system 2 consisting of a mixture of water vapor and gaseous mixture of hydrogen and oxygen in which H_2 and O_2 are present in the proportion of 2:1 and system 3 consisting of a mixture of water vapor and gaseous mixture of hydrogen and oxygen in which H_2 and O_2 are present in the proportion of 1:1



Illustrating the definition of pure substance

Illustration of definition of pure substance

- Analyzing system 1, we can observe that proportion of H_2 and O_2 is identical whether the sample is taken from liquid water and water vapor. Similarly, chemical combination ratio of H_2 and O_2 is identical for both liquid water and water vapor. Hence, system 1 satisfies both conditions (a) and (b) and can be regarded as a pure substance as long as its homogeneity does not change with time (i.e. satisfies condition (c)).
- Analyzing system 2, we observe that the portions of H_2 and O_2 will be the same whether we take a sample from the water vapor or from the gaseous mixture of H_2 and O_2 . So it is homogeneous in composition. But if we go on reducing the sample size from the gaseous mixture we can get a single molecule of H_2 or O_2 whereas in case of sample from water vapor the smallest possible sample size is a single water molecule thus, not homogeneous in chemical aggregation.

Illustration of definition of pure substance

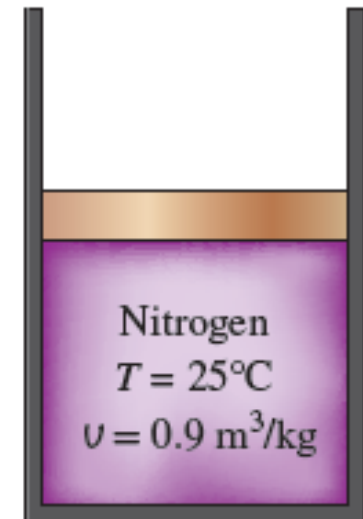
- Similarly, if we analyze system 3, we can observe that it is neither homogeneous in composition nor homogeneous in chemical aggregation. Hence, it is not a pure substance.

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State Postulate

- For a pure substance, we can determine all thermodynamic properties if two independent thermodynamic properties are given.
- Two properties are **independent** if one property can be varied while the other one is held constant.
- Temperature and specific volume, for example, are always independent properties, and together they can fix the state of a simple compressible system.
- Temperature and pressure, however, are independent properties for single-phase systems, but are dependent properties for multiphase systems.

The state of nitrogen is fixed by two independent, intensive properties.



State Postulate

- At sea level ($P = 1 \text{ atm}$), water boils at 100°C , but on a mountain top where the pressure is lower, water boils at a lower temperature. That is, $T = f(P)$ during a phase-change process.
- Thus, temperature and pressure are not sufficient to fix the state of a two-phase system.
- This special features of a pure substance is called **state postulate or two property rule** which is stated as:
- Two independent thermodynamic properties are sufficient to establish a stable thermodynamic state of a system composed of a pure substance.

$$T = T(P, V)$$

- Two independent intensive thermodynamic properties are sufficient to establish a stable thermodynamic state of a pure substance.

$$T = T(P, v)$$

Ideal Gas and Ideal Gas Relations

- A perfect gas or an ideal gas is defined as a gas having no forces of molecular attraction.
- A gas which follows the gas laws at all ranges of pressure and temperatures can be considered as an ideal gas but no such gas exists in nature.
- However, real gases tend to follow these laws at low pressure or high temperature. This is because the molecules are far apart at reduced pressures and elevated temperatures and the force of attraction between them tends to be small.
- The range of this low pressure and high temperature at which any gas behaves as an ideal gas is different for different gases.

Boyle's Law

Statement

- The volume of a given mass of a gas is inversely proportional to the pressure, provided that, the temperature remains constant.

$$V \propto \left(\frac{1}{P} \right)_T$$

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Charles's Law

This law is also known as Gay-Lussac's law and can be stated as

- The volume of a given mass of a gas is directly proportional to the temperature, provided that, the pressure remains constant.

$$V \propto (T)_P$$

- The pressure of a given mass of a gas is directly proportional to the temperature, provided that, the volume remains constant.

$$P \propto (T)_V$$

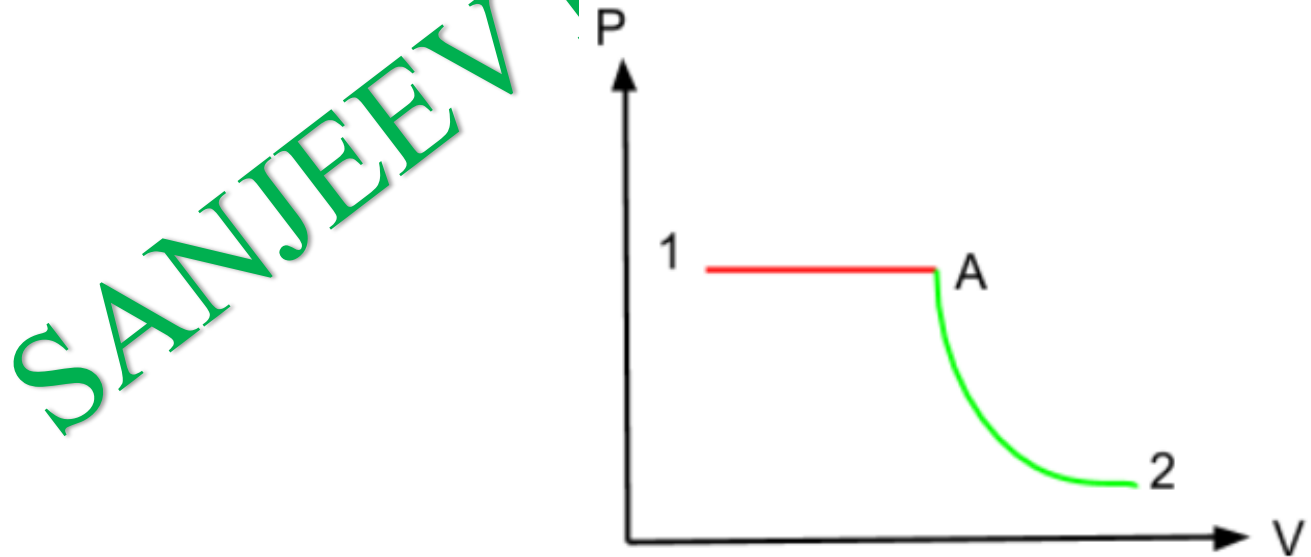
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Equation of State

- Any equation that relates the pressure, temperature, and specific volume of a substance is called an **equation of state**.
- Property relations that involve other properties of a substance at equilibrium states are also referred to as equations of state.
- There are several equations of state, some simple and others very complex.
- The simplest and best-known equation of state for substances in the gas phase is the ideal gas equation of state. This equation predicts the $P - v - T$ behavior of a gas quite accurately within some properly selected region.

More on Equation of State

- A relation between all three variables P , v and T may be obtained by combining Boyle's law and Charles' law.
- Consider unit mass of an ideal gas undergoing constant pressure process 1 – A followed by constant temperature process A – 2 as shown in figure.



Relation between P , v and T

More on Equation of State

Applying Charles's law for the process 1 – A

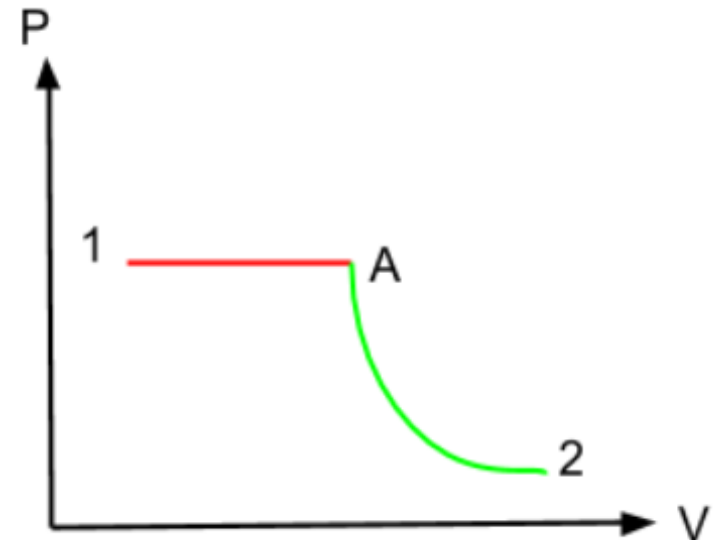
$$\frac{v_1}{v_A} = \frac{T_1}{T_A}$$

$$v_A = \frac{T_A}{T_1} v_1 = \frac{T_2}{T_1} v_1$$

Applying Boyle's law for the process A – 2

$$\frac{v_2}{v_A} = \frac{P_A}{P_2}$$

$$v_A = \frac{P_2}{P_A} v_2 = \frac{P_2}{P_1} v_2$$



More on Equation of State

Equating the relations for v_A , we get

$$\frac{P_1 v_1}{T_1} = \frac{P_2 v_2}{T_2}$$

Since state 1 and 2 are selected arbitrarily,

$$\frac{P_1 v_1}{T_1} = \frac{P_2 v_2}{T_2} = \frac{Pv}{T} = R$$

$$Pv = RT$$

where R is a constant for a given gas. The constant R is called the **characteristics gas constant** and has unit of J/kgK.

- For a system consisting of a mass of m kg of gas, we can write

$$PV = mRT$$

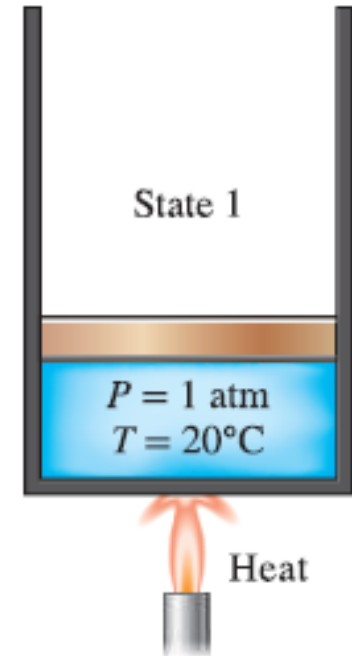
Gas Constant for Common Gases

Gas	R, J/kgK	C _p , J/kgK	C _v , J/kgK	γ
Air	287	1005	718	1.40
Argon	208.1	520.3	312.2	1.667
Helium	2076.9	5192.5	3115.6	1.667
Nitrogen	296.8	1039.8	743	1.40
Oxygen	259.8	917.8	658	1.395

SAMPLE

Two Phase (Liquid and Vapor) Systems: Saturation Curve on T – v Diagram

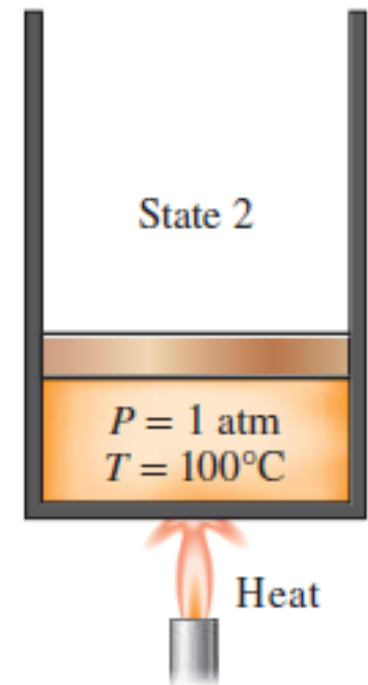
- Consider a frictionless piston cylinder device containing liquid water at 20°C and 1 atm pressure specified by state 1. Under these conditions, water exists in the liquid phase, and it is called a **compressed liquid**, or a **subcooled liquid**.
- The pressure in the cylinder remains constant at 1 atm during this process since it depends on the outside barometric pressure and the weight of the piston, both of which are constant.



At 1 atm and 20°C, water exists in the liquid phase (compressed liquid).

Saturation Curve on T – v Diagram

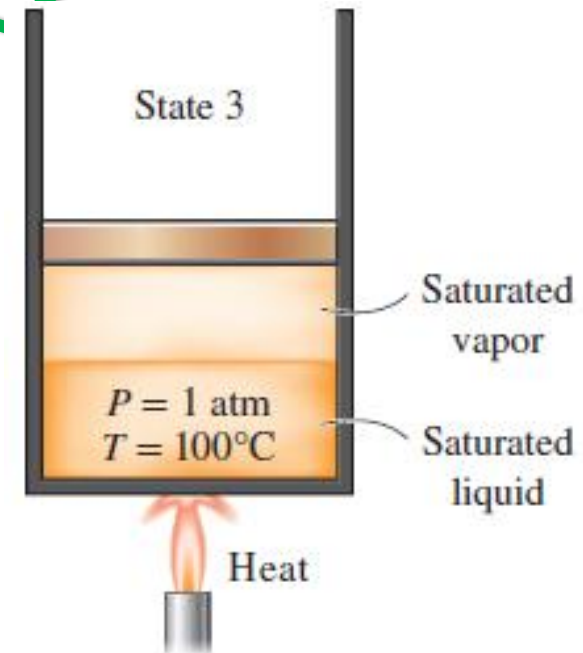
- As more heat is transferred, the temperature keeps rising with slight increment of specific volume which is represented by **line 1 – 2**. This trend continues until its temperature reaches 100°C , state 2.
- At this point water is still a liquid, but any heat addition will cause some of the liquid to vaporize. That is, a phase – change process from liquid to vapor is about to take place.
- A liquid that is about to vaporize is called a **saturated liquid**. Therefore, state 2 is a saturated liquid state.



At 1 atm pressure and 100°C , water exists as a liquid that is ready to vaporize (saturated liquid).

Saturation Curve on T – v Diagram

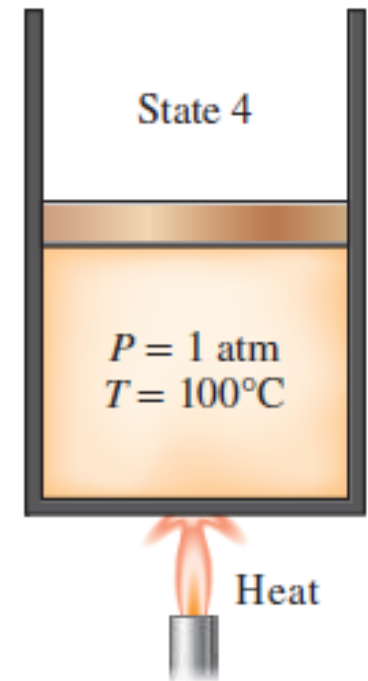
- Once boiling starts, the temperature stops rising until the liquid is completely vaporized. That is, the temperature will remain constant during the entire phase – change process if the pressure is held constant and specific volume of substance increases due to intermolecular expansion.
- Midway about the vaporization line, state 3, the cylinder contains equal amounts of liquid and vapor.



As more heat is transferred, part of the saturated liquid vaporizes (saturated liquid – vapor mixture).

Saturation Curve on T – v Diagram

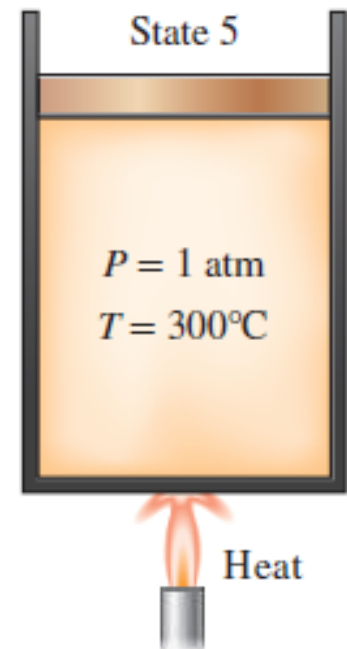
- As we continue transferring heat, the vaporization process continues until the last drop of liquid is vaporized, state 4 (after the absorption of complete latent heat). This evaporation process is represented by **line 2 – 4**. Any heat loss from this vapor will cause some of the vapor to condense (phase change from vapor to liquid).
- A vapor that is about to condense is called a **saturated vapor**. Therefore, state 4 is a saturated vapor state.
- A substance at states between 2 and 4 is referred to as a **saturated liquid–vapor mixture** since the liquid and vapor phases coexist in equilibrium at these states.



At 1 atm pressure, the temperature remains constant at 100°C until the last drop of liquid is vaporized (saturated vapor).

Saturation Curve on T – v Diagram

- Once the phase – change process is completed, we are back to a single phase region again (this time vapor), and further transfer of heat results in an increase in both the temperature and the specific volume shown by **line 4 – 5**.
- At state 5, the temperature of the vapor is, let us say, 300°C ; and if we transfer some heat from the vapor, the temperature may drop somewhat but no condensation will take place as long as the temperature remains above 100°C (for $P = 1\text{ atm}$).



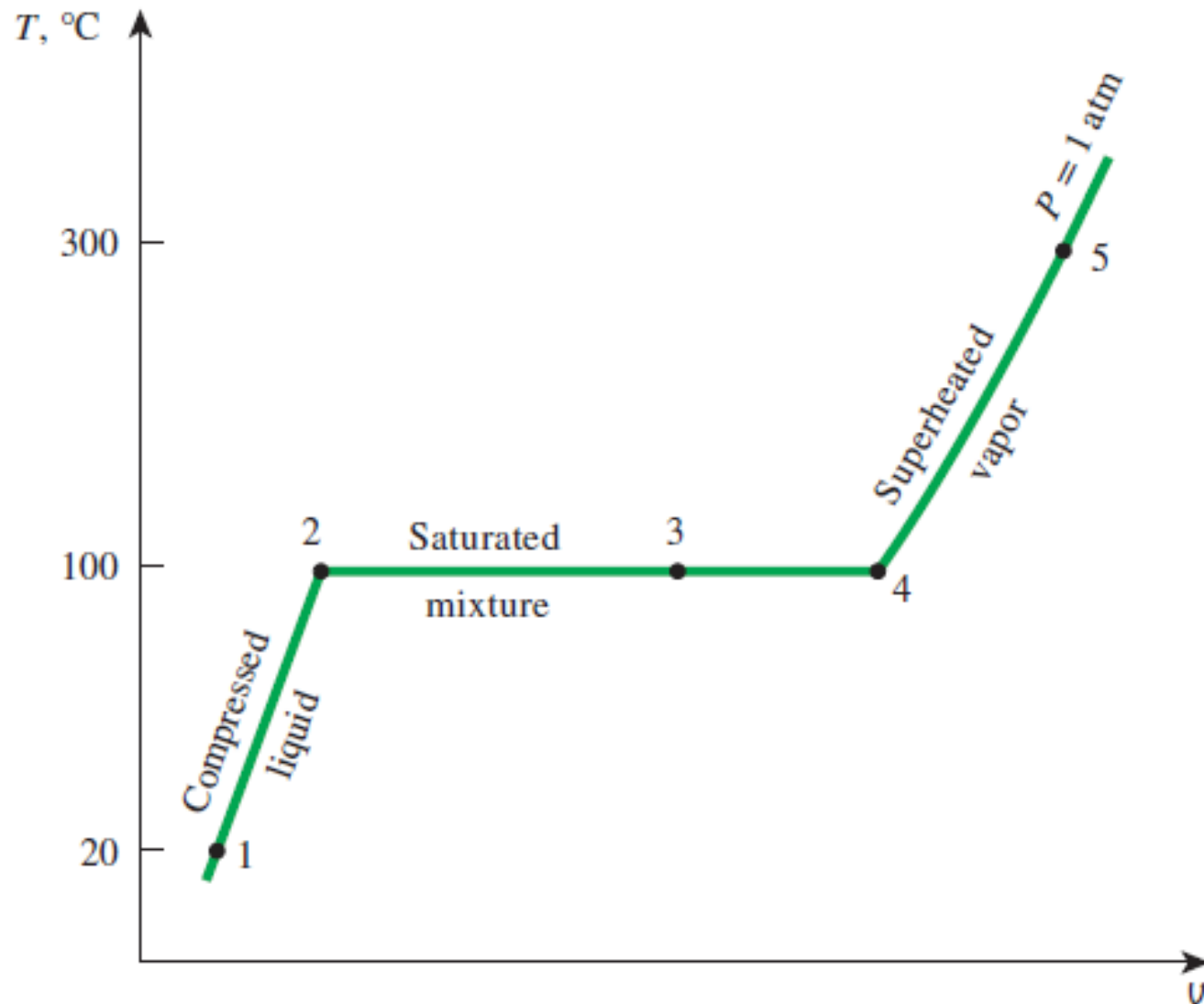
As more heat is transferred, the temperature of the vapor starts to rise (superheated vapor).

Saturation Curve on $T - v$ Diagram

- A vapor that is not about to condense (i.e., not a saturated vapor) is called a **superheated vapor**.
- Therefore, water at state 5 is a superheated vapor.
- This constant pressure phase change process is illustrated on a $T - v$ diagram.

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Saturation Curve on $T - v$ Diagram



$T - v$ diagram for the heating process of water at constant pressure

Saturation Curve on T – v Diagram

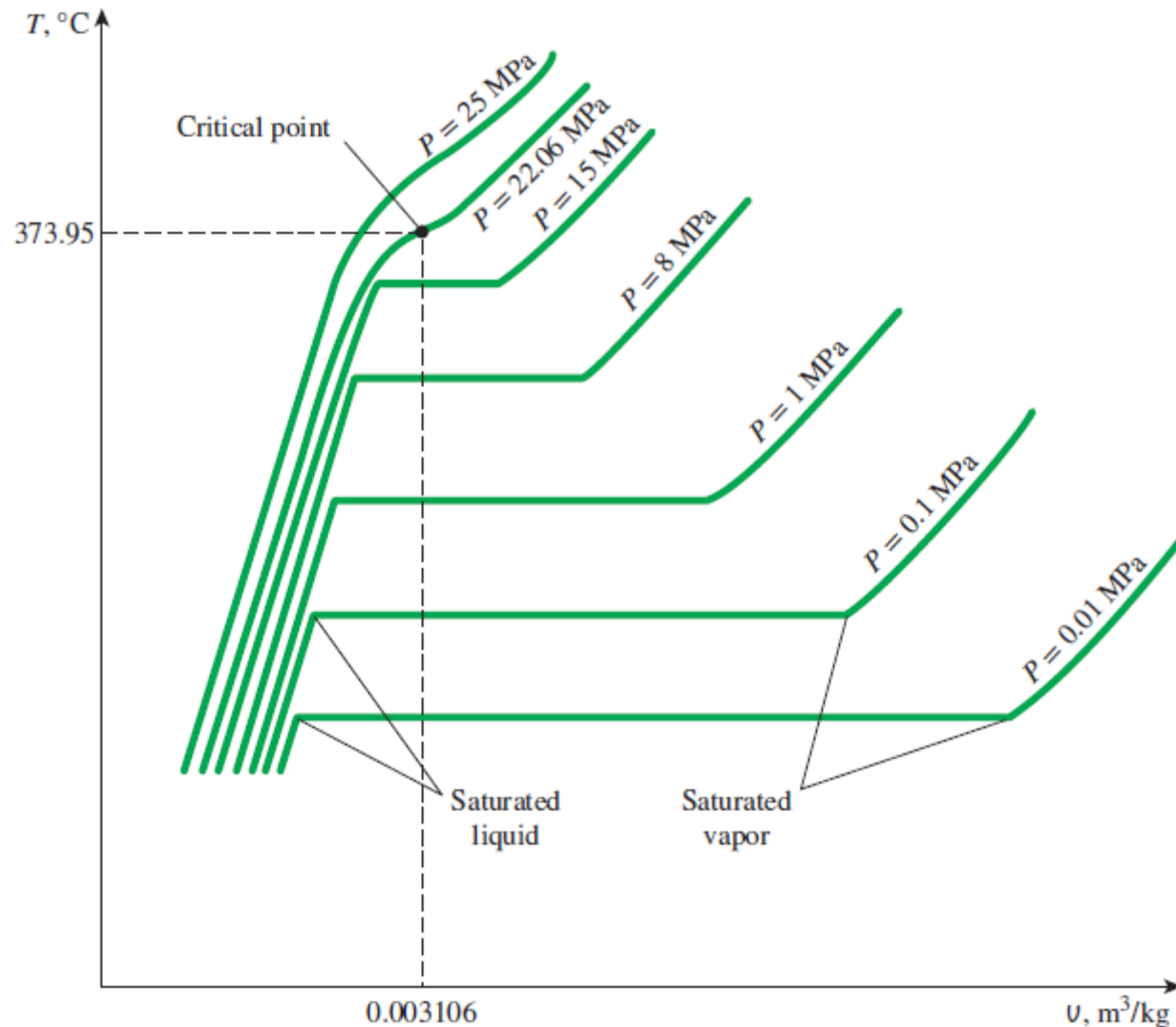
- Let us add weights on top of the piston until the pressure inside the cylinder reaches 1 MPa. At this pressure, water has a somewhat smaller specific volume than it does at 1 atm pressure. As heat is transferred to the water at this new pressure, the process follows a path that looks very much like the process path at 1 atm pressure, but there are some noticeable differences.
- First, water starts boiling at a much higher temperature (179.9°C) at this pressure.
- Second, the specific volume of the saturated liquid is larger and the specific volume of the saturated vapor is smaller than the corresponding values at 1 atm pressure. That is, the horizontal line that connects the saturated liquid and saturated vapor states is much shorter.

Saturation Curve on T – v Diagram

- As the pressure is increased further, evaporation takes place relatively at higher temperatures and the length of line 2 – 4 goes on decreasing.
- Thus, the saturation line continues to shrink, and it becomes a point when the pressure reaches 22.06 MPa for the case of water.
- This point is called the **critical point**, and it is defined as the point at which the saturated liquid and saturated vapor states are identical.

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Saturation Curve on T – v Diagram



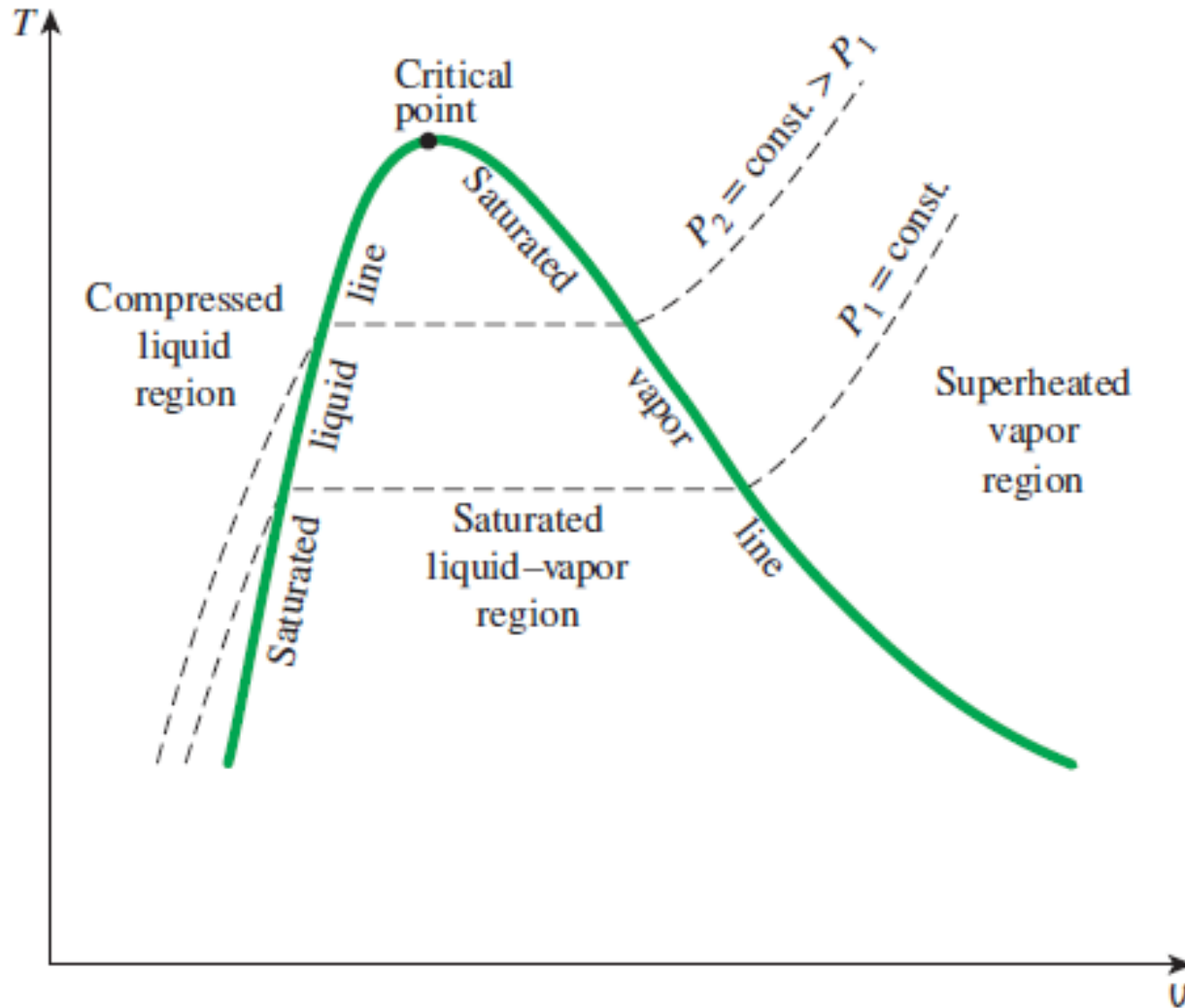
$T - v$ diagram of constant pressure phase change processes of a pure substance at various pressures

Saturation Curve on T – v Diagram

- The saturated liquid states can be connected by a line called the **saturated liquid line**, and saturated vapor states can be connected by another line, called the **saturated vapor line**. The entire curve so formed is called saturation curve.
- These two lines meet at the critical point, forming a dome.
- All the compressed liquid states are located in the region to the left of the saturated liquid line, called the **compressed liquid region**.
- All the superheated vapor states are located to the right of the saturated vapor line, called the **superheated vapor region**.
- In these two regions, the substance exists in a single phase, a liquid or a vapor.
- All the states that involve both phases in equilibrium are located under the dome, called the **saturated liquid–vapor mixture region**, or the **wet region**.

Saturation Curve on T – v Diagram

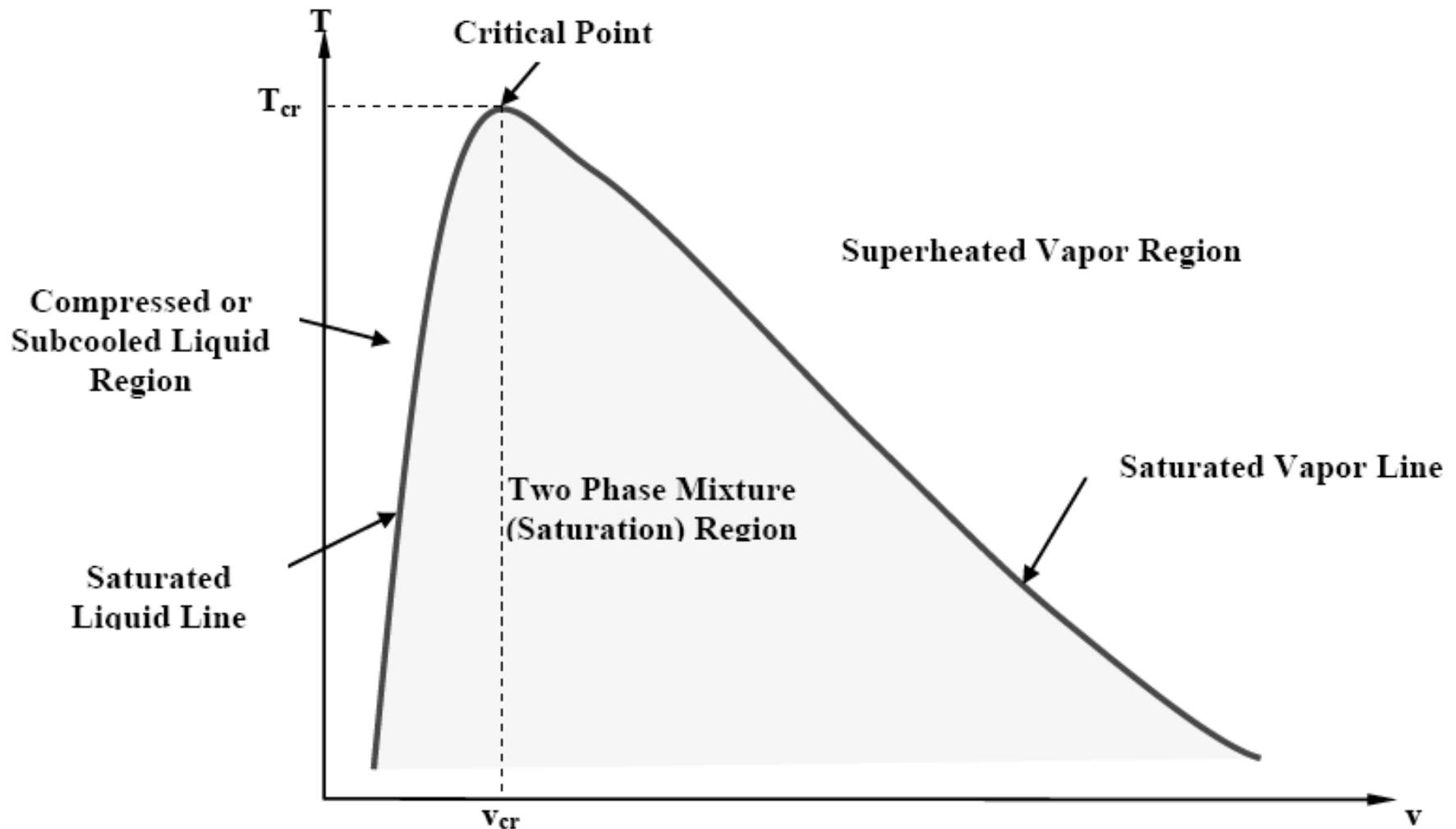
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Saturation curve for two phase mixture on T – v diagram

Saturation Curve on T – v Diagram

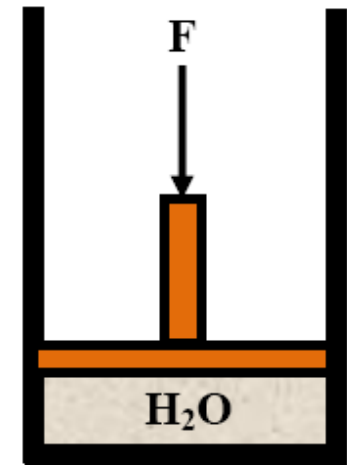
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Saturation curve for two phase mixture on T – v diagram

Two Phase (Liquid and Vapor) Systems: Saturation Curve on P – v Diagram

- Consider a piston cylinder arrangement containing water at compressed liquid state, say at 5 atm and 100°C. The system is placed in contact with a source which has a temperature slightly higher than that of H₂O inside the cylinder.
- If the external force F is reduced slowly such that temperature is maintained constant at 100°C, its pressure decreases and there will be slight increment in its specific volume, as shown by line AB.



The pressure in a piston–cylinder device can be reduced by reducing the weight of the piston.

Saturation Curve on P – v Diagram

- This trend continues until its pressure reduces to 1 atm, that is, it reaches saturated liquid state B.
- When saturated liquid take heat from the source it gets evaporated and its specific volume increases until it reaches to saturated vapor state C, while its pressure and temperature remains constant as shown by line BC.
- If pressure is further reduced from state C, its specific volume increases as shown by the curve CD.
- If we repeat similar expansion process at higher temperatures, evaporation takes place correspondingly at higher pressure.

Page 33 and 34

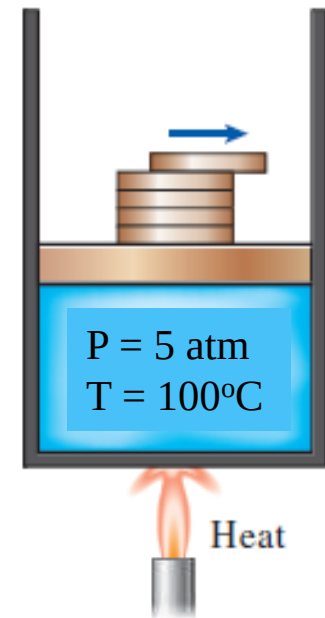
OR

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Same Explanation

Two Phase (Liquid and Vapor) Systems: Saturation Curve on P – v Diagram

- Consider a piston–cylinder device that contains liquid water at 5 atm and 100°C. Water at this state exists as a compressed liquid. Now the weights on top of the piston are removed one by one so that the pressure inside the cylinder decreases gradually. The water is allowed to exchange heat with the surroundings so its temperature remains constant. As the pressure decreases, the volume of the water increases slightly.

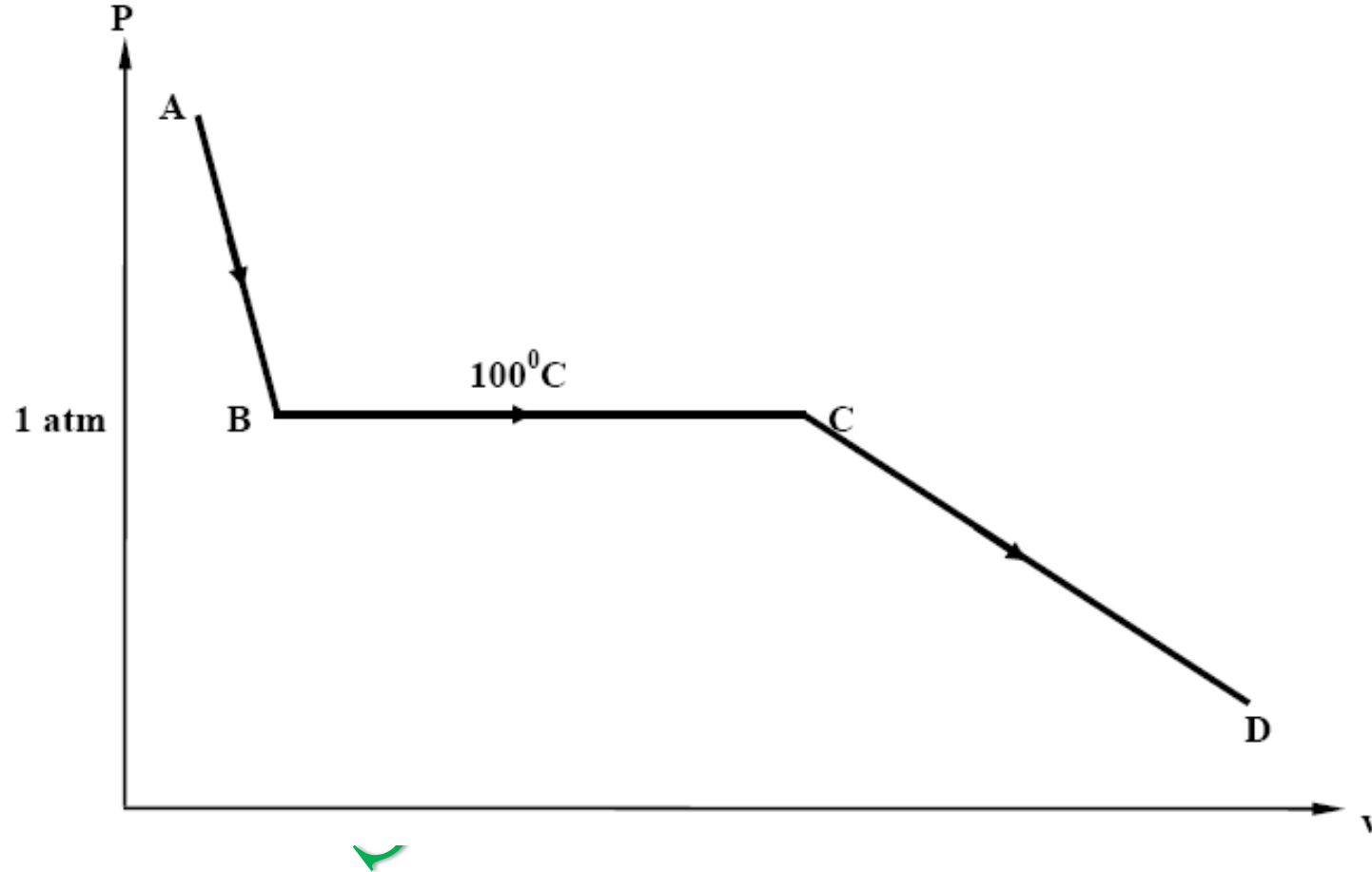


The pressure in a piston–cylinder device can be reduced by reducing the weight of the piston.

Saturation Curve on $P - v$ Diagram

- When the pressure reaches the saturation-pressure value at the specified temperature (1 atm), the water starts to boil.
- During this vaporization process, both the temperature and the pressure remain constant, but the specific volume increases. Once the last drop of liquid is vaporized, further reduction in pressure results in a further increase in specific volume.
- Notice that during the phase-change process, we did not remove any weights. Doing so would cause the pressure and therefore the temperature to drop [since $T_{\text{sat}} = f(P_{\text{sat}})$], and the process would no longer be isothermal.
- When the process is repeated for other temperatures, similar paths are obtained for the phase-change processes. Connecting the saturated liquid and the saturated vapor states by a curve, we obtain the $P - v$ diagram of a pure substance.

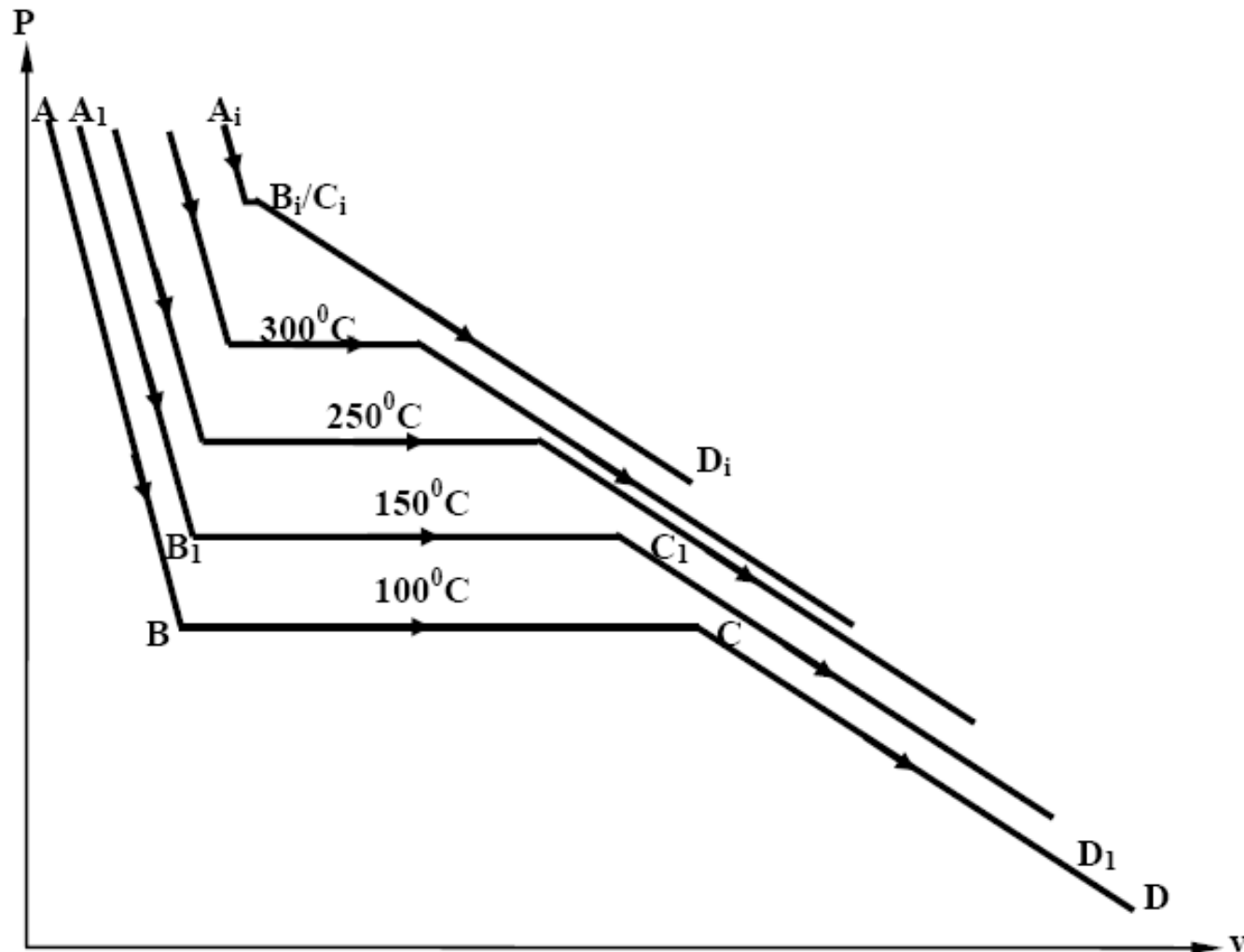
Saturation Curve on $P - v$ Diagram



$P - v$ diagram for the heating process of water at constant temperature

Saturation Curve on P – v Diagram

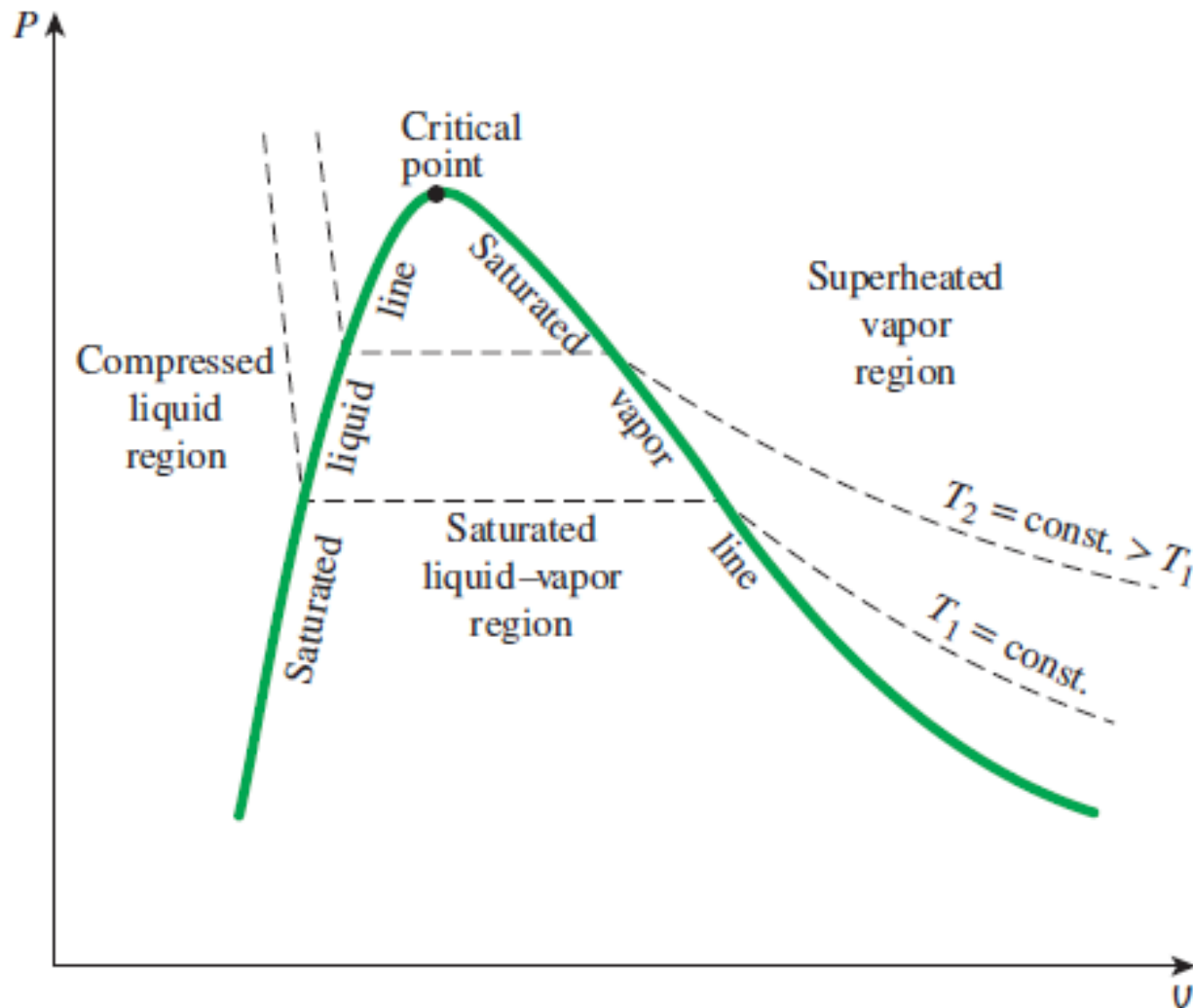
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$P - v$ diagram of constant temperature phase change processes of a pure substance at various temperature

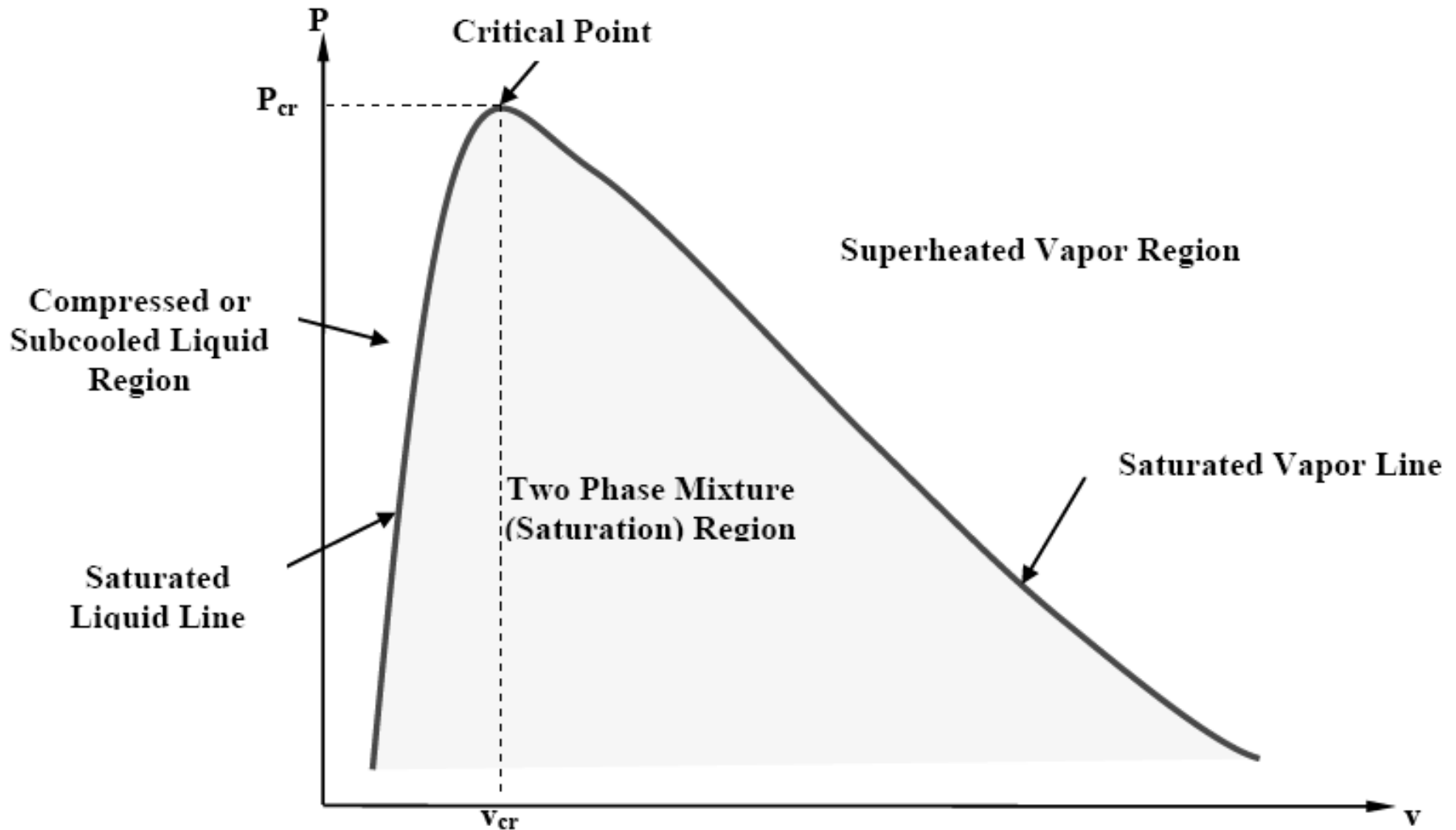
Saturation Curve on P – v Diagram

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Saturation curve for two phase mixture on P – v diagram

Saturation Curve on P – v Diagram



Saturation curve for two phase mixture on P – v diagram

Terms used in Two – Phase Mixture System

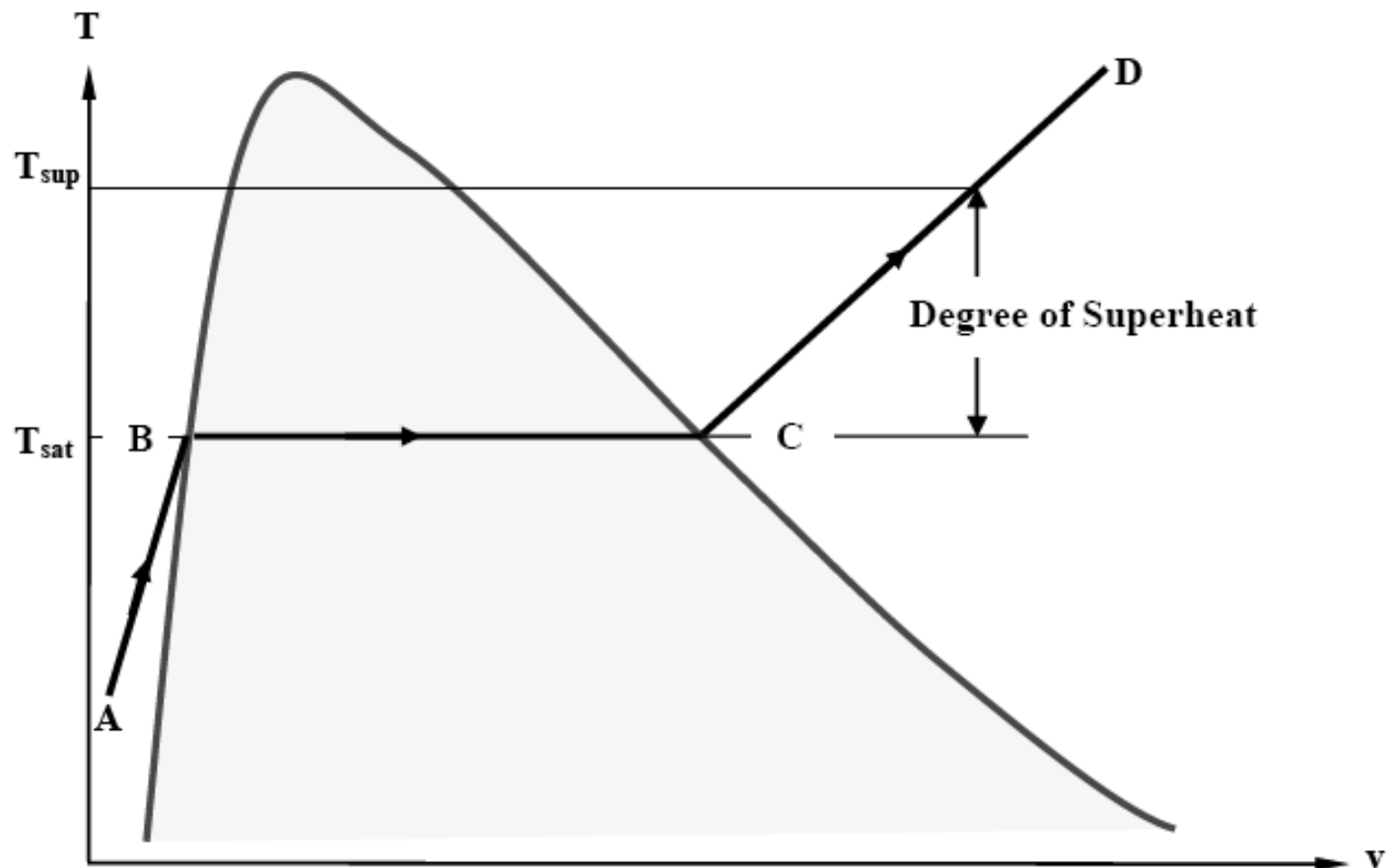
- **Saturation temperature** – For a given pressure, the temperature at which phase change (liquid to vapor or vapor to liquid) occurs is called a **saturation temperature**. For example, if we heat water at 1 atm, evaporation takes place at 100°C, i.e., saturation temperature for 1 atm is 100°C.
- **Saturation Pressure** – For a given temperature, the pressure at which phase change (liquid to vapor or vapor to liquid) occurs is called a **saturation pressure**. For example, if we expand water at 100°C, evaporation takes place 1 atm at, i.e., saturation pressure for 100°C is 1 atm.
- **Saturated Liquid** – The state of a substance at which evaporation just starts (during heating) or condensation just completes (during cooling) is called the **saturated liquid**.

Terms used in Two – Phase Mixture System

- **Saturated Vapor** – The state of a substance at which evaporation just completes (during heating) or condensation just starts (during cooling) is called the **saturated vapor**.
- **Compressed Liquid or Subcooled Liquid** – When saturated liquid is further cooled or compressed, it is called a **compressed liquid or subcooled liquid**.
- **Superheated Vapor** – When saturated vapor is further heated or expanded, it is called a **superheated vapor**.

Terms used in Two – Phase Mixture System

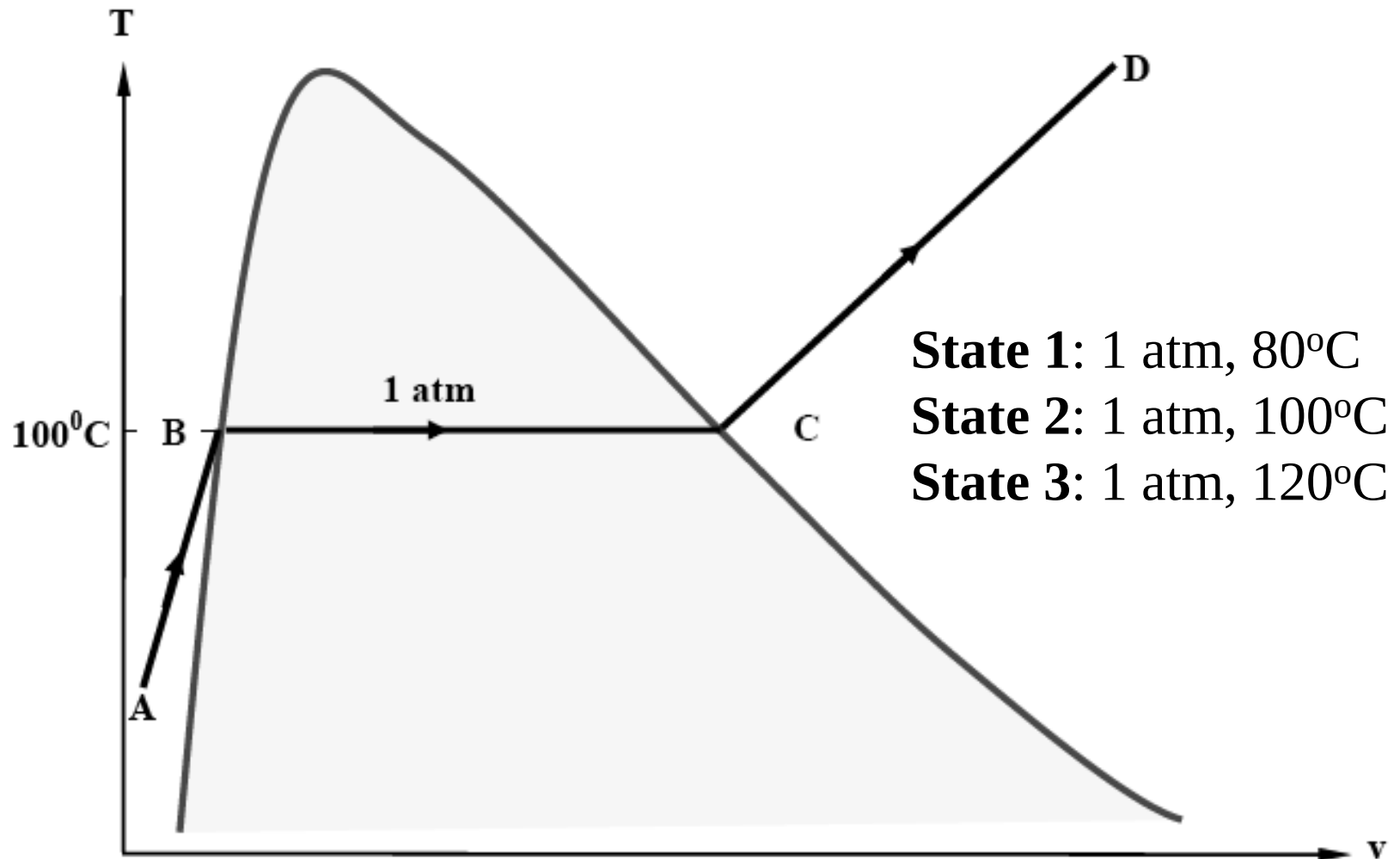
- Degree of Superheat** – The difference in temperature of a superheated vapor and the corresponding saturation temperature is called degree of superheat.



Terms used in Two – Phase Mixture System

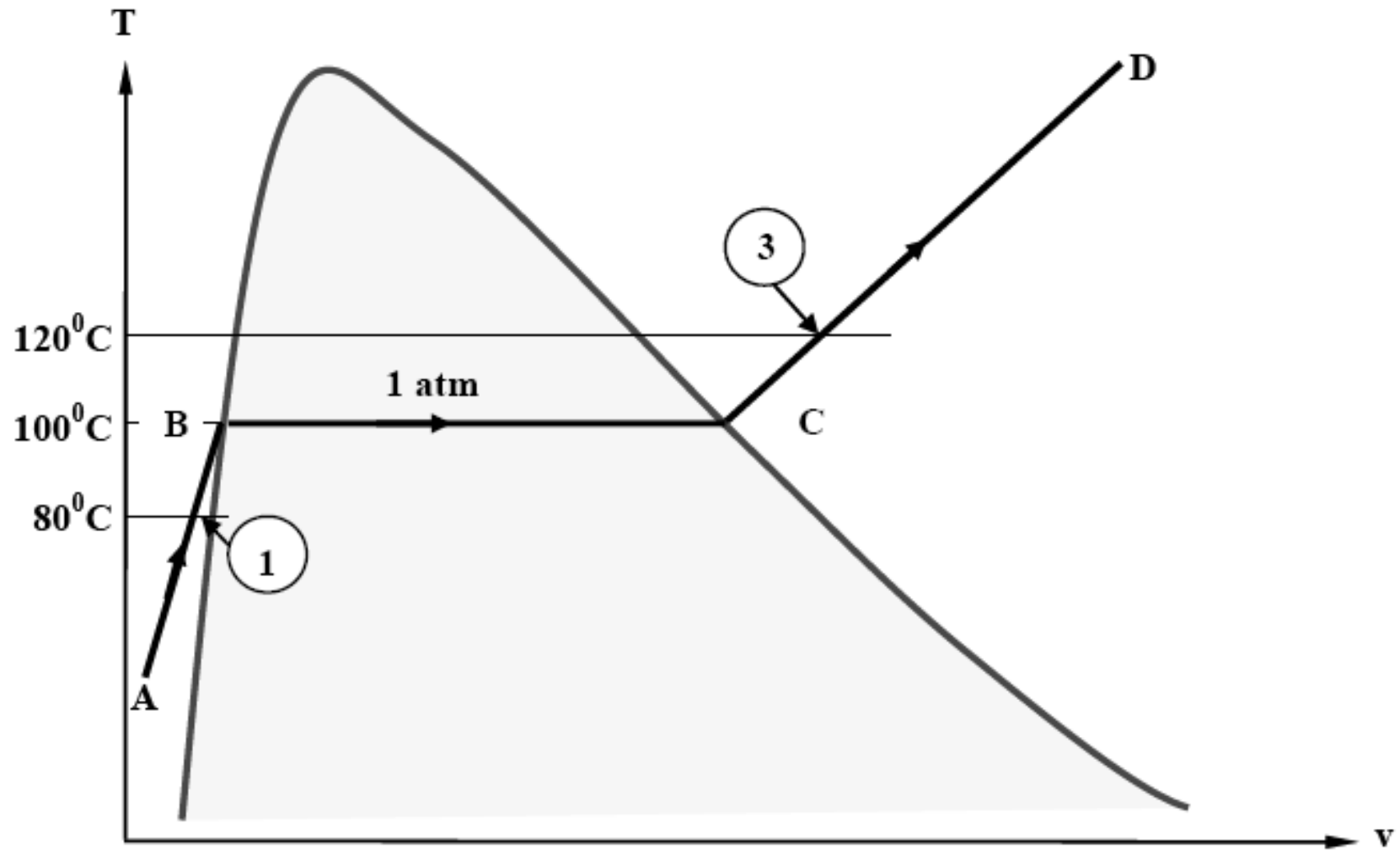
- **Critical Point** – The state of a substance at which saturated liquid and saturated vapor have identical properties is called a **critical point**. The pressure and temperature corresponding to critical point are called **critical pressure** and **critical temperature** respectively.
- For H_2O , critical pressure and critical temperature are **22.055 MPa** and **373°C**. If we H_2O at **22.055 MPa** it suddenly gets converted into saturated vapor from saturated liquid when temperature reaches **373°C**.

Properties of a Two – Phase Mixture



Saturation curve with an isobaric line

Properties of a Two – Phase Mixture



Locating states on $T - v$ with reference to saturation lines

Properties of a Two – Phase Mixture

For a given pressure,

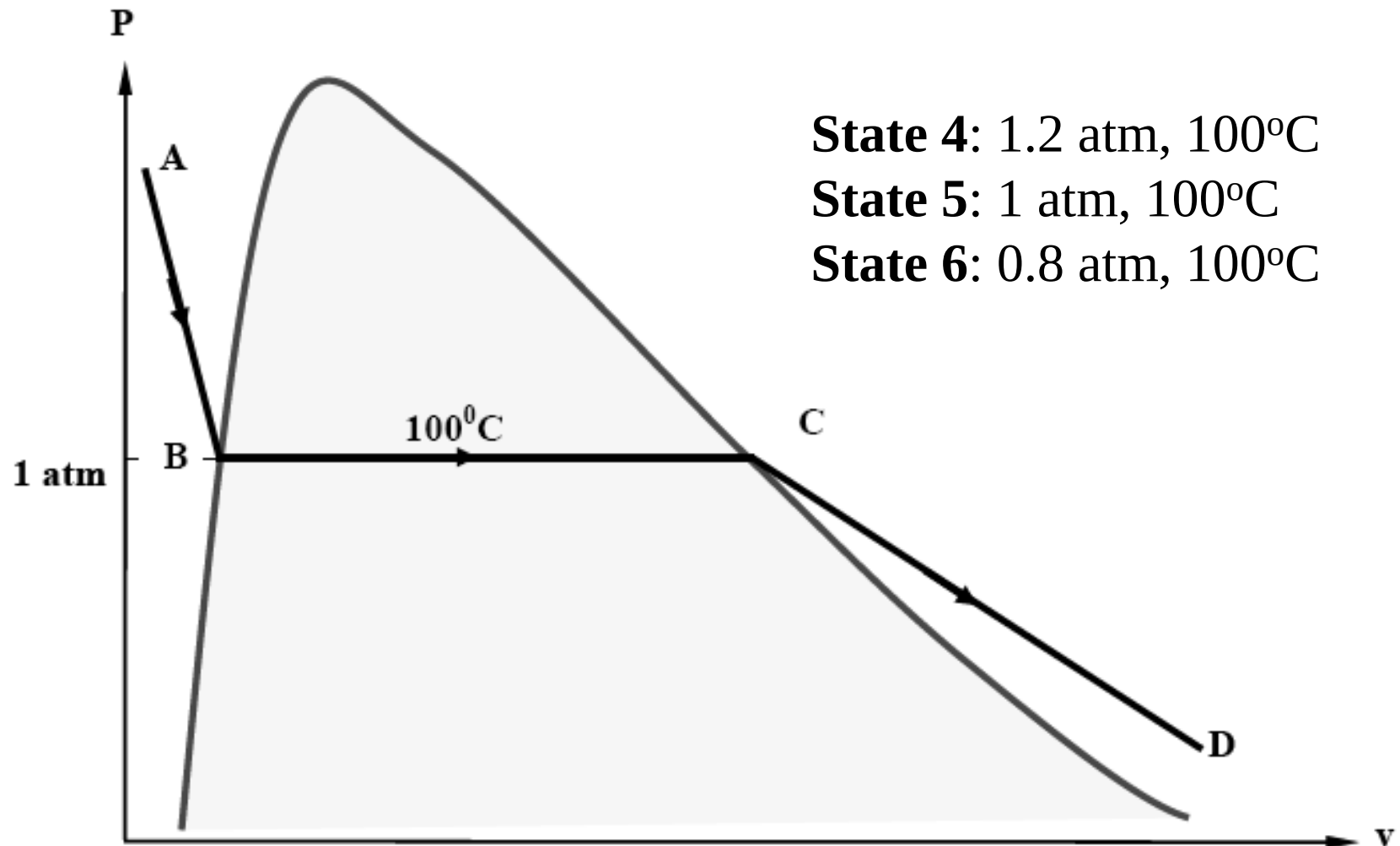
- If temperature is less than the saturation temperature, its state will be a compressed or sub-cooled liquid.
- If temperature is higher than the saturation temperature, its state will be a superheated vapor.

$T < T_{sat}$, its state will be a compressed or sub-cooled liquid.

$T > T_{sat}$, its state will be a superheated vapor.

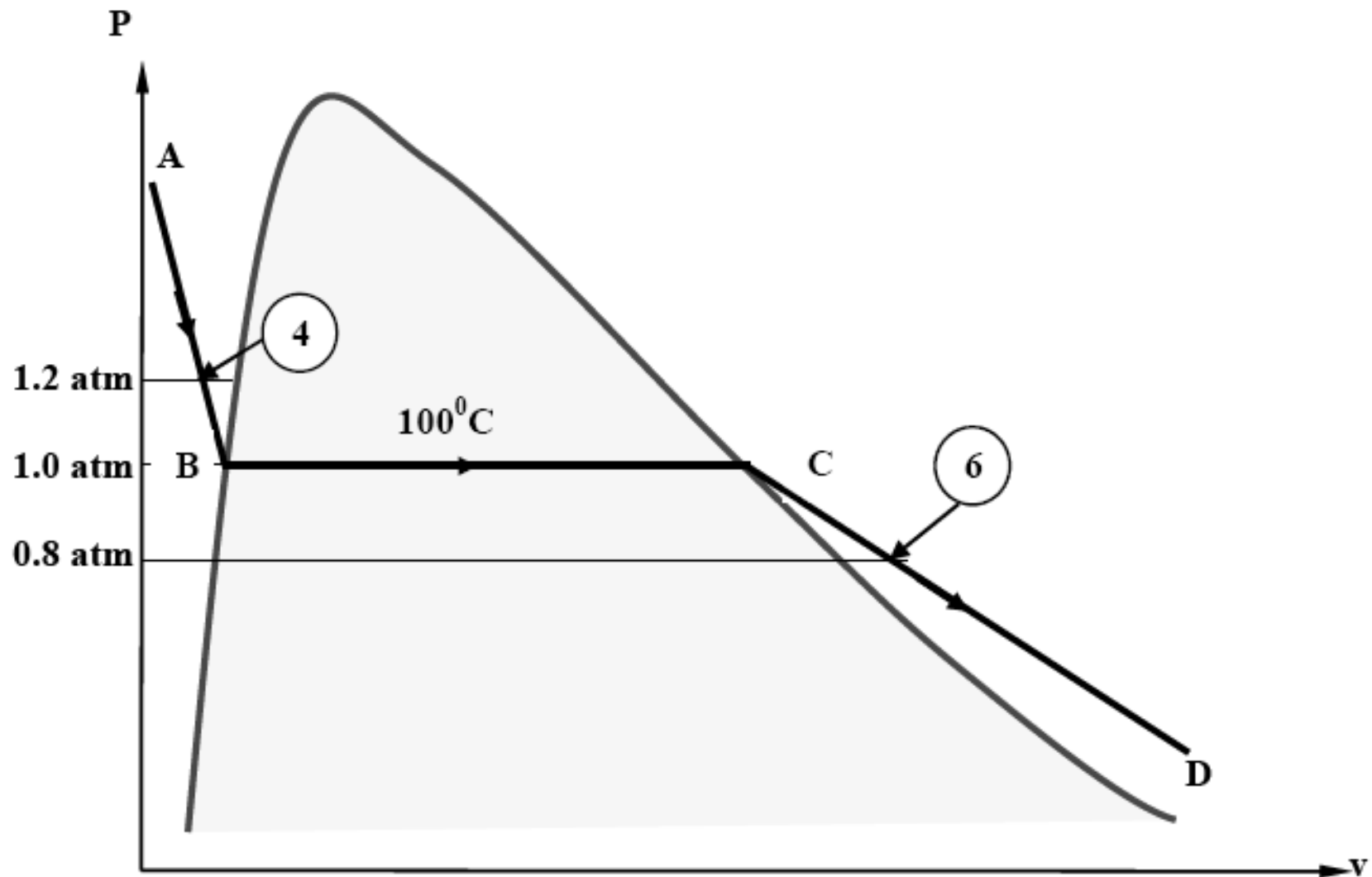
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Properties of a Two – Phase Mixture



Saturation curve with an isothermal line

Properties of a Two – Phase Mixture



Locating states on P – v with reference to saturation lines

Properties of a Two – Phase Mixture

For a given temperature,

- If pressure is higher than the saturation pressure, its state will be a compressed or sub-cooled liquid.
- If pressure is less than the saturation pressure, its state will be a superheated vapor.

$P > P_{sat}$, its state will be a compressed or sub-cooled liquid.

$P < P_{sat}$, its state will be a superheated vapor.

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Quality

- From the above examples, we can conclude that states 1,3, 4 and 6 can be fixed with given pressure and temperature. But states 2 and 5 cannot be located with identical given parameters. We can compare this difference with reference to state postulate.
- For compressed liquid or superheated vapor state, pressure and temperature are independent, hence we can fix the states with the given pressure and temperature.
- But for two phase mixture (saturation) region, pressure and temperature are no more independent, and we cannot fix the state with dependent pressure and temperature.
- To fix the state within the saturation region, quality is necessary.

Quality

- **Quality** is defined as the ratio of mass of saturated vapor and the total mass of the two phase mixture.
- It is also called dryness fraction and is denoted by x .

$$x = \frac{m_g}{m} = \frac{m_g}{m_l + m_g}$$

where m_g is the mass of saturated vapor
 m_l is the mass of saturated liquid, and
 m is the total mass of the total phase mixture.

- For saturated liquid state, $m = m_l$ or $m_g = 0$

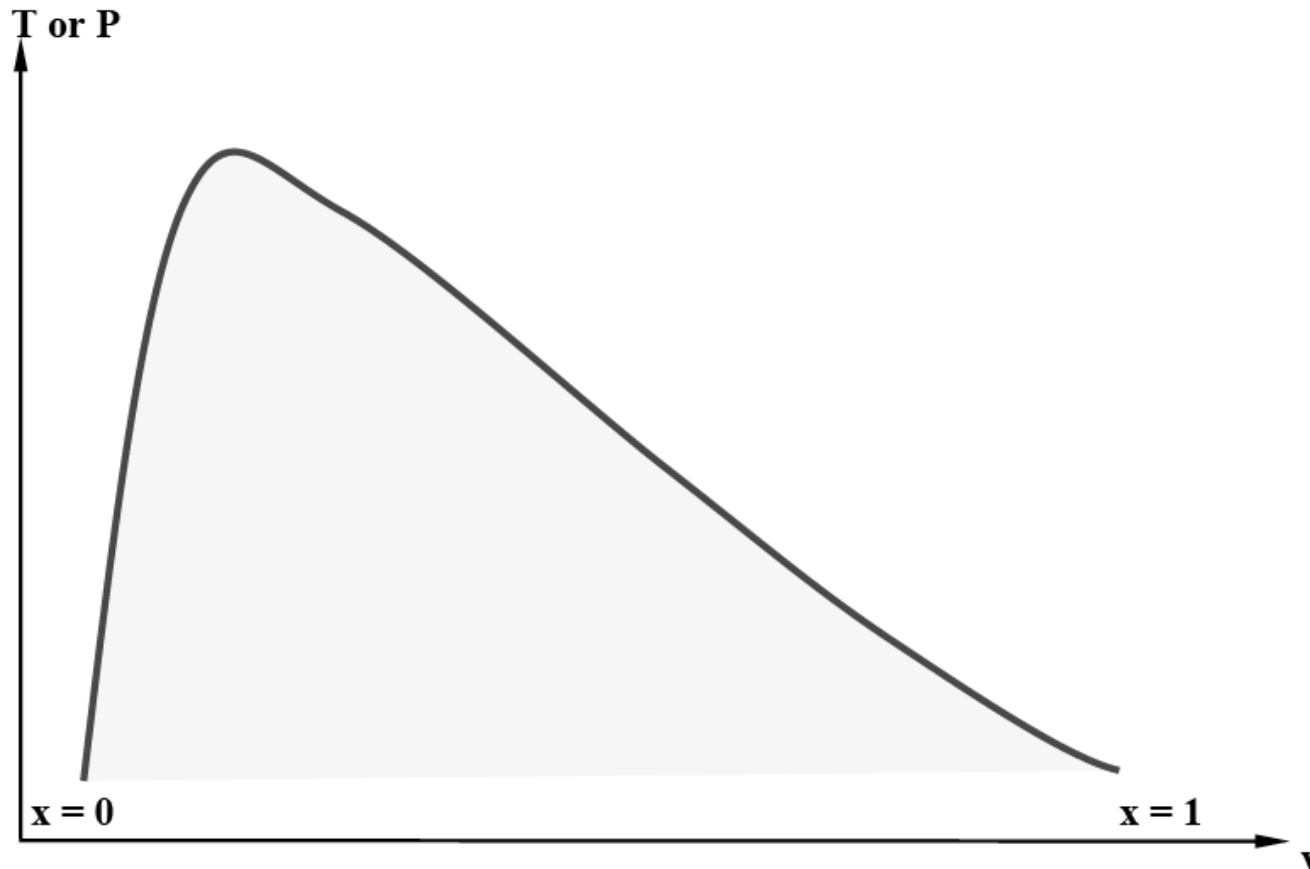
$$x_{sat \text{ liquid}} = 0$$

- For saturated vapor state, $m = m_g$ or $m_l = 0$

$$x_{sat \text{ vapor}} = 1$$

Quality

Hence, quality varies from 0 to 1 ($0 \leq x \leq 1$) within the saturation region.



Limiting values of quality

Moisture Content

- To fix the state within the saturation region we can also use the property called moisture content.
- Moisture content is defined as the ratio of mass of saturated liquid and the total mass of the two phase mixture.
- It is denoted by y .

$$y = \frac{m_l}{m} = \frac{m_l}{m_l + m_g} = 1 - x$$

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Specific Properties of a Two – Phase Mixture

- Specific properties of the two-phase mixture can be expressed in term of quality. For example, specific volume of a two phase mixture is given by the ratio of total volume of two – phase mixture and the total mass of the two phase mixture, i.e.

$$v = \frac{V}{m} = \frac{V_l}{m} + \frac{V_g}{m}$$

where V_l is the volume of saturated liquid, and V_g is the volume of saturated vapor.

$$v = \frac{V_l}{m_l} \frac{m_l}{m} + \frac{V_g}{m_g} \frac{m_g}{m} = v_l(1 - x) + v_g x$$

where v_l is the specific volume of saturated liquid, and v_g is the specific volume of saturated vapor.

Specific Properties of a Two – Phase Mixture

$$v = v_l(1 - x) + v_g x = v_l + x(v_g - v_l)$$

$$v = v_l + x v_{lg}$$

where $v_{lg} = v_g - v_l$, is the change in specific volume during evaporation.

- Other specific properties of the two-phase mixture can be expressed in the similar manner as

Specific internal energy

$$u = u_l + x u_{lg}$$

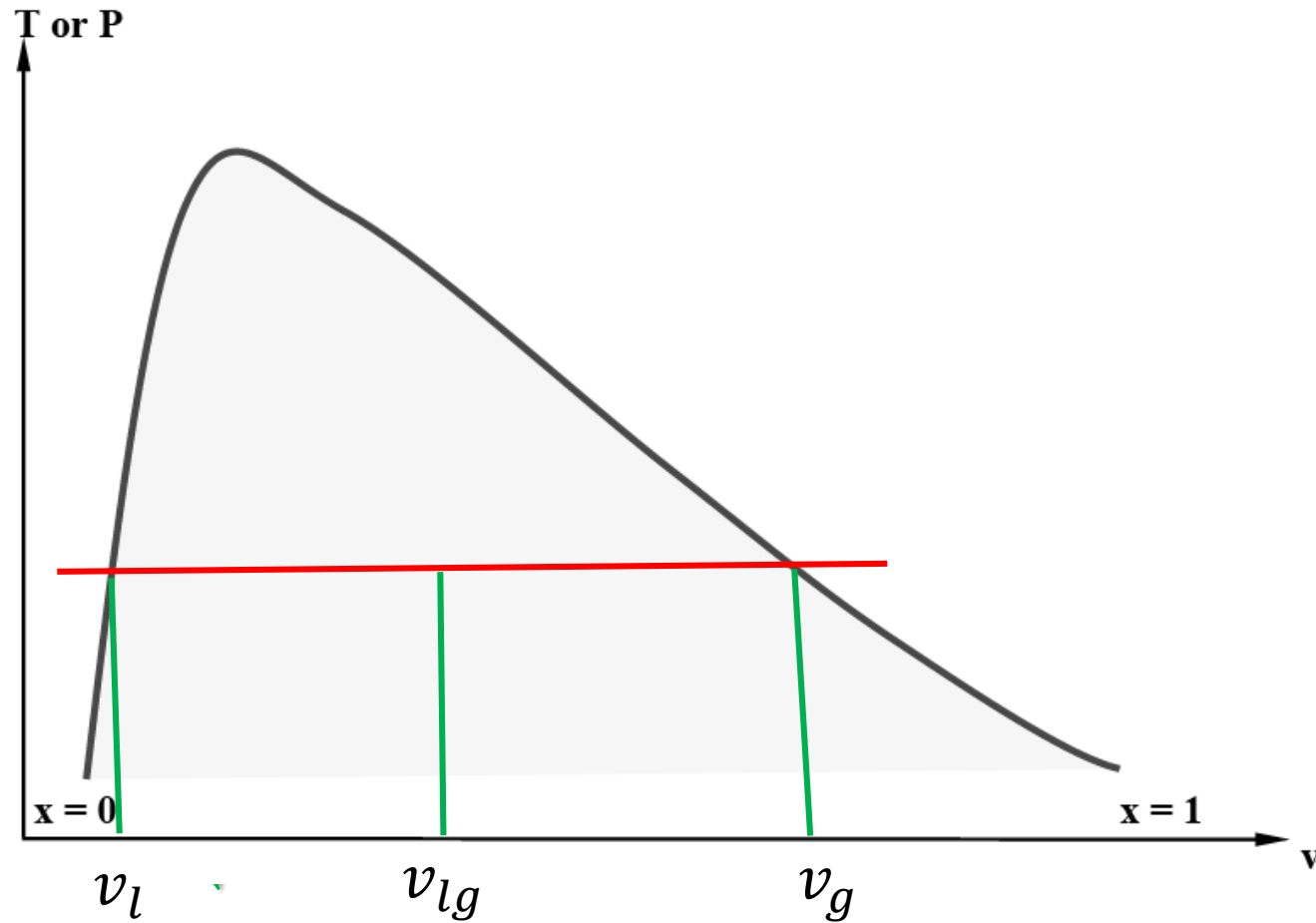
Specific enthalpy

$$h = h_l + x h_{lg}$$

Specific entropy

$$s = s_l + x s_{lg}$$

Properties of a Two – Phase Mixture



Properties of a Two – Phase Mixture

For a given specific volume,

- If specific volume is lesser than specific volume of saturated liquid, its state will be a compressed or sub-cooled liquid.
- If specific volume is higher than specific volume of saturated vapor, its state will be a superheated vapor.
- If the value lies between these two values, its state will be two phase mixture.

$v < v_l$, the state is compressed / sub-cooled liquid.

$v > v_g$, the state is superheated vapor

$v_l < v < v_g$, the state is two phase mixture.

Development of Property Data

- Properties of a substance at within saturation region or near saturation region cannot be determined using the mathematical equations (e.g. ideal gas equation). Hence, properties under such conditions are determined experimentally and presented in the form of tables or charts.
- Property tables can be used to determine the property of a substance for a particular state.
- Property charts can be used to determine the properties for a particular state as well as process can also be plotted.

Concept Application 3.1

State	P, kPa	T, °C	x, %	v, m ³ /kg	h, kJ/kg	Degree of Superheat
1	200	200				
2	250		80			
3	300			0.5		
4		150	60			
5		200		0.1		
6		250		0.5951		
7	1000				2000	
8		350			2563.5	
9	600			0.001101		
10	2000			0.1757		
11	5000			0.001206		
12		500			3445.4	
13	8000	295.04				

State	P, kPa	T, °C	x, %	v, m ³ /kg	h, kJ/kg	Degree of Superheat, °C
1	200	200	×	1.0803	2870.0	79.76
2	250	127.44	80	0.575227	2280.53	×
3	300	133.56	82.495	0.5	2346.544	×
4	475.72	150	60	0.23617	1900.78	×
5	1553.6	200	78.385	0.1	2373.135	×
6	400	250	×	0.5951	2963.6	106.36
7	1000	179.92	61.402	0.11982	2000	×
8	16520	350	100	0.008812	2563.5	0
9	600	158.86	0	0.001101	670.71	×
10	2000	500	×	0.1757	3467.7	287.58
11	5000	230	×	0.001206	990.43	×
12	4000	500	×	0.08642	3445.4	249.61
13	8000	295.04	—	—	—	—

A 0.3 m^3 rigid vessel contains 5 kg of water at 150 kPa. Determine:

- (a) the temperature,
- (b) the mass of each phase, and
- (c) the specific enthalpy

Solution:

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continued

Given, Volume of vessel (V) = 0.3 m^3

Mass of water (m) = 5 kg

Pressure (P) = 150 kPa

Specific volume of water (v) = $\frac{V}{m} = \frac{0.3}{5} = 0.06 \text{ m}^3/\text{kg}$

Referring to Table A2.1

T	P	v_l	v_{lg}	v_g	h_l	h_{lg}	h_g
$^{\circ}\text{C}$	kPa	m^3/kg	m^3/kg	m^3/kg	kJ/kg	kJ/kg	kJ/kg
145	415.29	0.001085	0.4453	0.4464	610.75	2129.4	2740.2
150	475.72	0.001090	0.3918	0.3929	632.32	2114.1	2746.4
155	542.99	0.001096	0.3457	0.3468	653.95	2098.4	2752.3

Here, $v_l < v < v_g$, hence it is a two phase mixture. Quality of mixture is given as

continued

$$x = \frac{v - v_l}{v_{lg}} = \frac{0.06 - 0.001053}{1.1584} = 0.05088$$

a) Temperature of mixture is given as

$$T = T_{\text{sat}} (150 \text{ kPa}) = 111.38^\circ \text{C}$$

b) Mass of saturated vapor (m_g) = $x \times m$

$$= 0.05088 \times 5 = 0.2544 \text{ kg}$$

Mass of saturated liquid (m_l) = $m - m_g$

$$= 5 - 0.2544 = 4.7456 \text{ kg}$$

c) Specific enthalpy of mixture is given as

$$h = h_l + xh_g = 467.18 + 0.05088 \times 2226.2$$

$$= 580.4491 \text{ kJ/kg}$$

Concept Application 3.3

A rigid vessel having a volume of 0.02 m^3 , initially contains water at its critical state. The vessel is cooled until its pressure drops to 2000 kPa. Determine:

- (a) the mass of H_2O present in the vessel,
- (b) the quality at final state, and
- (c) the mass of saturated liquid water and saturated water vapor at the final state. (TU 2068 Bhadra)

Solution:

continued

Given, Volume of vessel (V) = 0.02 m^3

State 1: Critical point

State 2: $P_2 = 2000 \text{ kPa}$

Specific volume at initial state, $v_1 = v_{\text{cr}} = 0.00311 \text{ m}^3/\text{kg}$

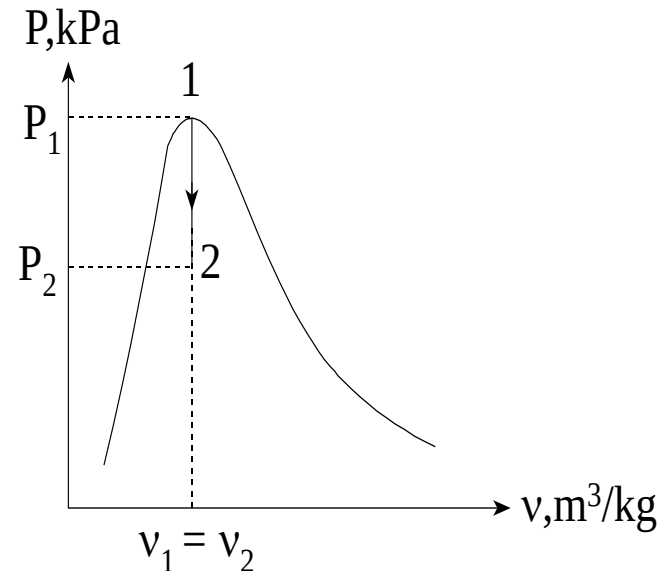
Process 1 – 2: constant volume cooling process

$$\text{Mass of H}_2\text{O, } m = \frac{V}{v} = \frac{0.02}{0.00311} = 6.431 \text{ kg}$$

Specific volume of H_2O at final state,

$$v_2 = v_1 = 0.00311 \text{ m}^3/\text{kg}$$

Referring to Table A2.1,



continued

P	T	v_l	v_{lg}	v_g
kPa	$^{\circ}\text{C}$	m^3/kg	m^3/kg	m^3/kg
1900	209.84	0.001172	0.1035	0.1047
2000	212.42	0.001177	0.09841	0.09959
2250	218.45	0.001187	0.08753	0.08872
22,055	373.98	0.00311	-	0.00311

Then quality of H_2O at final state is given as

$$x_2 = \frac{v_2 - v_l}{v_{lg}} = \frac{0.00311 - 0.001177}{0.09841} = 0.01964$$

Masses of saturated liquid and saturated vapor at final states are given as

$$(m_g)_2 = x_2 m = 0.01964 \times 6.431 = 0.1263 \text{ kg}$$

$$(m_l)_2 = m - (m_g)_2 = 6.431 - 0.1263 = 6.3047 \text{ kg}$$

Sample Problem 3.1

Steam contained in a closed container initially at a pressure of 2 MPa and a temperature of 250°C. The temperature drops as a result of heat transfer to the surroundings until the temperature reaches 80°C. Determine:

- the pressure at which the condensation first occurs,
- the pressure and quality at final state, and
- the percentage of volume occupied by the saturated liquid at the final state. (TU 2074 Ashwin)

Solution:

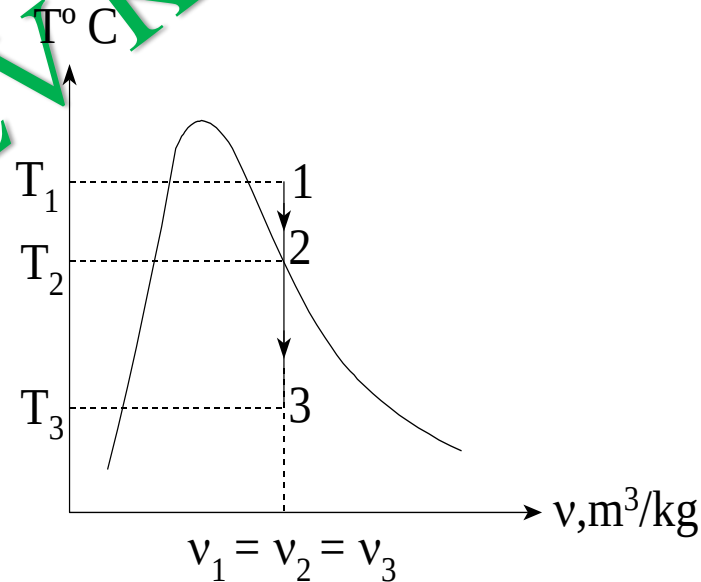
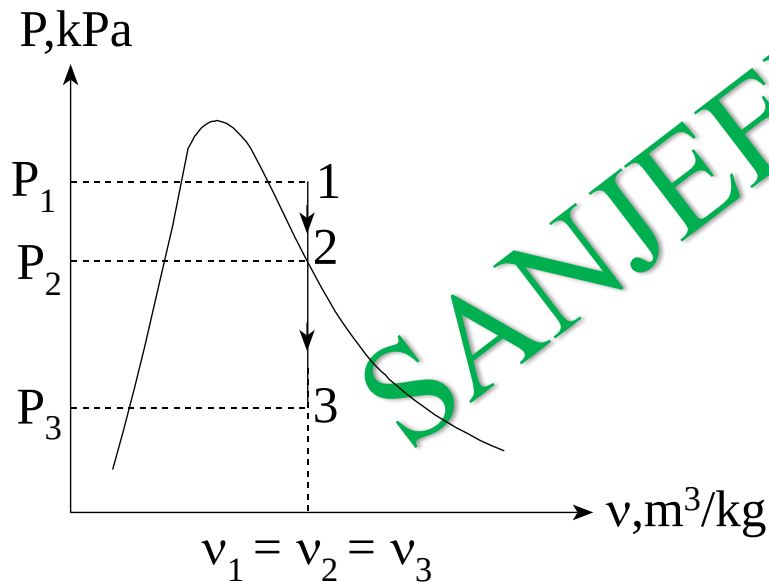
continued

Given, Initial state: $P_1 = 2 \text{ MPa} = 2000 \text{ kPa}$, $T_1 = 250^\circ \text{C}$

Final state: $T_{\text{final}} = 80^\circ \text{C}$

Now, referring to the Table A2.1, $T_{\text{sat}} (2000 \text{ kPa}) = 212.42^\circ \text{C}$.
Here, $T_1 > T_{\text{sat}}$, hence it is a superheated steam.

Now, referring to Table A2.4, $v_1 = 0.1114 \text{ m}^3/\text{kg}$



continued

From $P - v$, the condensation first occurs at P_2 kPa.

Process: constant volume cooling process

State 2: $v_2 = 0.1114 \text{ m}^3/\text{kg}$

Referring to Table A2.1, specific volumes of saturated vapor which includes specific volume $0.1114 \text{ m}^3/\text{kg}$ and corresponding pressure are listed as

$P, \text{ kPa}$	$v_g, \text{ m}^3/\text{kg}$	
1700	0.1167	(a)
P_2	0.1114	
1800	0.1104	(b)

continued

Then, applying linear interpolation for pressure,

$$P_2 - P_a = \frac{P_b - P_a}{(v_g)_b - (v_g)_a} [v_2 - (v_g)_a]$$

$$\therefore P_2 = P_a + \frac{P_b - P_a}{(v_g)_b - (v_g)_a} [v_2 - (v_g)_a]$$

$$= 1700 + \frac{1800 - 1700}{0.1104 - 0.1167} (0.1114 - 0.1167) = 1784.127 \text{ kPa}$$

The pressure at which the condensation first occurs, P_2

$$= 1784.127 \text{ kPa}$$

State 3: $v_3 = 0.1114 \text{ m}^3/\text{kg}$

Referring to Table A2.2,

$$v_l (80^\circ \text{C}) = 0.001029 \text{ m}^3/\text{kg}, v_{lg} (80^\circ \text{C}) = 3.4078 \text{ m}^3/\text{kg}, v_g (80^\circ \text{C}) = 3.4088 \text{ m}^3/\text{kg}, P_{\text{sat}} (80^\circ \text{C}) = 47.373 \text{ kPa}.$$

Here, $v_l < v < v_g$, hence it is a two phase mixture.

continued

Quality of mixture at state 3 is given as

$$x_3 = \frac{v_3 - v_l}{v_{lg}} = \frac{0.1114 - 0.001029}{3.4078} = 0.0324$$

Pressure at final state is given as

$$P_3 = P_{\text{sat}} (80^\circ \text{C}) = 47.373 \text{ kPa}$$

Then, mass fraction of liquid $\left(\frac{m_l}{m}\right) = y_3 = 1 - x_3 = 1 - 0.0324 = 0.9675$

Percentage of volume occupied by the saturated liquid at final state

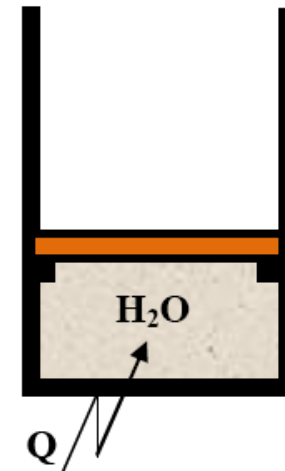
$$= \frac{V_l}{V} = \frac{v_l \times m_l}{v_3 \times m} = \frac{0.001029}{0.1114} \times 0.9675 \times 100 \% = 0.894 \%$$

Sample Problem 3.2

A piston cylinder device shown in figure below contains 0.2 kg of a mixture of saturated liquid water and saturated water vapor at a temperature of 50°C and a volume of 0.03 m^3 . The mass of the piston resting on the stops is 50 kg and the cross sectional area of the piston is 12.2625 cm^2 . The atmospheric pressure is 100 kPa. Heat is transferred until it becomes saturated vapor. Sketch the process on $P - v$ and $T - v$ diagrams and determine:

- (a) the temperature at which the piston just leaves the stops,
- (b) the final pressure, and
- (c) the total work transfer. [Take $g = 9.81\text{ m/s}^2$] (TU 2069 Chaitra)

Solution:



continued

Given, Mass of water (m) = 0.2 kg

Initial state: $T_1 = 50^\circ \text{C}$, $V_1 = 0.03 \text{ m}^3$, Two phase mixture

Final state: Saturated vapor

Mass of piston (m_p) = 50 kg

Area of piston (A_p) = $12.2625 \text{ cm}^2 = 12.2625 \times 10^{-4} \text{ m}^2$

Specific volume at initial state (v_1) = $\frac{V_1}{m} = \frac{0.03}{0.2} = 0.15 \text{ m}^3/\text{kg}$

Referring to Table A2.2, $P_1 = P_{\text{sat}} (50^\circ \text{C}) = 12.344 \text{ kPa}$, $v_l (50^\circ \text{C}) = 0.001012 \text{ m}^3/\text{kg}$, $v_g (50^\circ \text{C}) = 12.037 \text{ m}^3/\text{kg}$, $v_{lg} (50^\circ \text{C}) = 12.036 \text{ m}^3/\text{kg}$

$\therefore$ Pressure at state 1 is given as

$P_1 = P_{\text{sat}} (50^\circ \text{C}) = 12.344 \text{ kPa}$

continued

Free Body Diagram

$$P_{\text{abs}} = P_{\text{atm}} + \frac{W}{A_p}$$

$$\therefore P_2 = 100 + \frac{50 \times 9.81}{12.2625 \times 10^{-4}} \times 10^{-3}$$

Thus, pressure required to lift the piston (P_{lift}) = 500 kPa

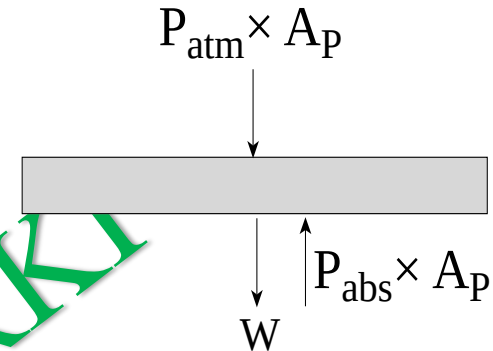
Process 1 – 2: Constant volume heating process

State 2: $P_2 = 500$ kPa, $V_2 = 0.03$ m³, $v_2 = 0.15$ m³/kg

Referring to Table A2.1, $T_{\text{sat}} (500 \text{ kPa}) = 151.87^\circ\text{C}$, $v_l (500 \text{ kPa}) = 0.001093$ m³/kg, $v_{lg} (500 \text{ kPa}) = 0.3738$ m³/kg, $v_g (500 \text{ kPa}) = 0.3749$ m³/kg. Here, $v_l < v < v_g$, hence it is a two phase mixture.

Process 2 – 3: Constant pressure heating process

State 3: $P_3 = 500$ kPa, Saturated vapor, $v_3 = 0.3749$ m³/kg,



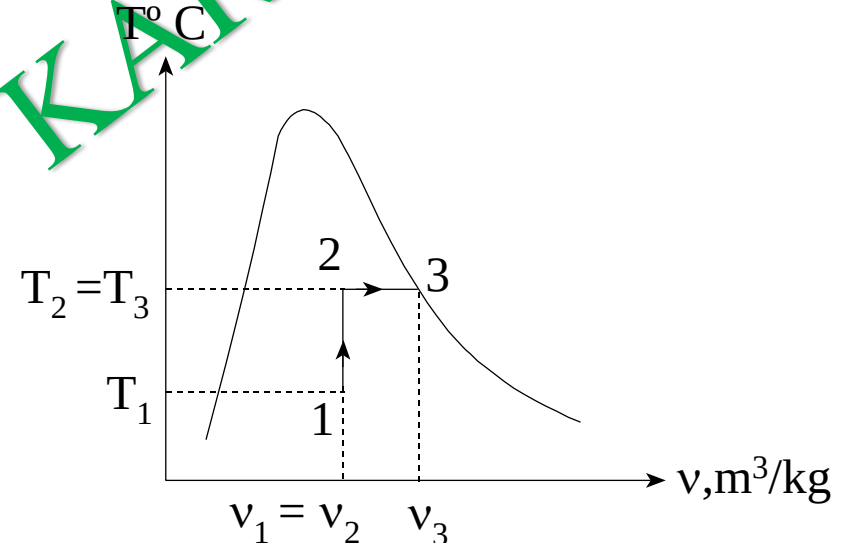
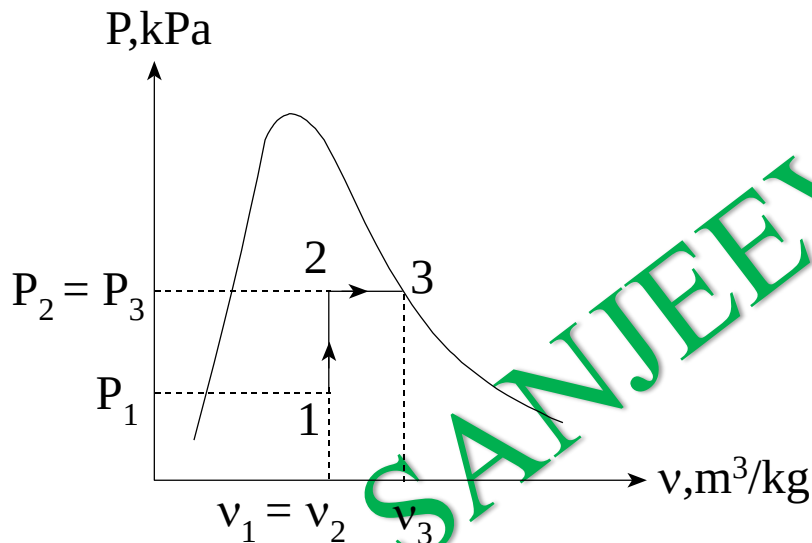
continued

The temperature at which piston just leaves the stops is given as

$$T_2 = T_{\text{sat}} (500\text{kPa}) = 151.87^\circ\text{C}$$

The final pressure is given as

$$P_3 = 500 \text{ kPa}$$



The total work transfer for the process is given as

$$\begin{aligned} W &= W_{12} + W_{23} = 0 + P_2 (V_3 - V_2) = mP_2 (v_3 - v_2) \\ &= 0.2 \times 500 (0.3749 - 0.15) = 22.49 \text{ kJ} \end{aligned}$$

Sample Problem 3.3

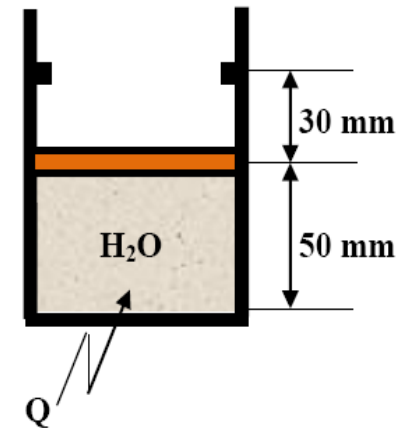
The frictionless piston shown in figure below has a mass of 20 kg and a cross sectional area of 78.48 cm^2 . Heat is added until the temperature reaches 400°C . If the quality of the H_2O at the initial state is 0.2, determine:

- the initial pressure,
- the mass of H_2O ,
- the quality of the system when the piston hits the stops,
- the final pressure, and
- the total work transfer.

[Take $P_{\text{atm}} = 100 \text{ kPa}$ and $g = 9.81 \text{ m/s}^2$]

(TU 2074 Bhadra)

Solution:



continued

Given, mass of piston (m_p) = 20 kg

Area of piston (A_p) = $78.48 \text{ cm}^2 = 78.48 \times 10^{-4} \text{ m}^2$

Initial state: $x_1 = 0.2$

Final state: $T_{\text{final}} = 400^\circ \text{C}$

Atmospheric pressure (P_{atm}) = 100 kPa

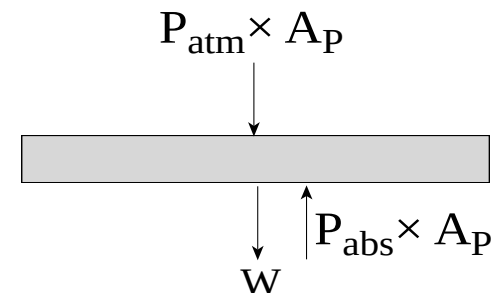
Initial volume of the system (V_1) = $A_p \times 0.05$
 $= 78.48 \times 10^{-4} \times 0.05 = 0.0003924 \text{ m}^3$

Final volume of the system (V_{final}) = $A_p \times 0.08$
 $= 78.48 \times 10^{-4} \times 0.08 = 0.00062784 \text{ m}^3$

Free Body Diagram

$$P_{\text{abs}} = P_{\text{atm}} + \frac{W}{A_p} = P_{\text{atm}} + \frac{m_p g}{A_p}$$

$$\therefore P_1 = 100 + \frac{20 \times 9.81 \times 10^{-3}}{78.48 \times 10^{-4}} = 125 \text{ kPa}$$



continued

Now, referring to Table A 2.1, $T_{\text{sat}} (125 \text{ kPa}) = 105.99^\circ\text{C}$, $v_l (125\text{kPa}) = 0.001048 \text{ m}^3/\text{kg}$, $v_g (125\text{kPa}) = 1.3752 \text{ m}^3/\text{kg}$, $v_{lg} (125 \text{ kPa}) = 1.3742 \text{ m}^3/\text{kg}$,

Specific volume at state 1 is given as

$$v_1 = v_l + x_1 v_{lg} = 0.001088 + 0.2 \times 1.3742 = 0.27589 \text{ m}^3/\text{kg}$$

$$\text{Then, mass of H}_2\text{O (m)} = \frac{V_1}{v_1} = \frac{0.0003924}{0.27589} = 0.001422 \text{ kg}$$

$$\therefore \text{Specific volume at final state (v}_{\text{final}}) = \frac{V_{\text{final}}}{m} = \frac{0.00062784}{0.001422} = 0.44142 \text{ m}^3/\text{kg}$$

Process 1 – 2: Constant pressure heating process

State 2: $P_2 = 125 \text{ kPa}$, $v_2 = 0.44142 \text{ m}^3/\text{kg}$

Here, $v_l < v_2 < v_g$, hence, it is a two phase mixture.

Quality of steam at state 2 is given as

$$x_2 = \frac{v_2 - v_l}{v_{lg}} = \frac{0.44142 - 0.001048}{1.3742} = 0.321$$

continued

And, temperature at state 2 is given as

$$T_2 = T_{\text{sat}} (125 \text{ kPa}) = 105.99^\circ \text{C}$$

Process 2 – 3: Constant volume heating process

State 3: $v_3 = 0.44142 \text{ m}^3/\text{kg}$, $T_3 = 400^\circ \text{C}$

Referring to Table A2.2, $T_{\text{cr}} = 373.98^\circ \text{C}$. Here, $T_3 > T_{\text{cr}}$, hence it is a superheated vapor. Then, referring to Table A2.4, specific volumes of superheated vapor which includes the specific volume $0.44142 \text{ m}^3/\text{kg}$ and corresponding pressure for 400°C are listed as

P, kPa	$v_g, \text{m}^3/\text{kg}$	
600	0.5137	(a)
P_2	0.44142	
800	0.3843	(b)

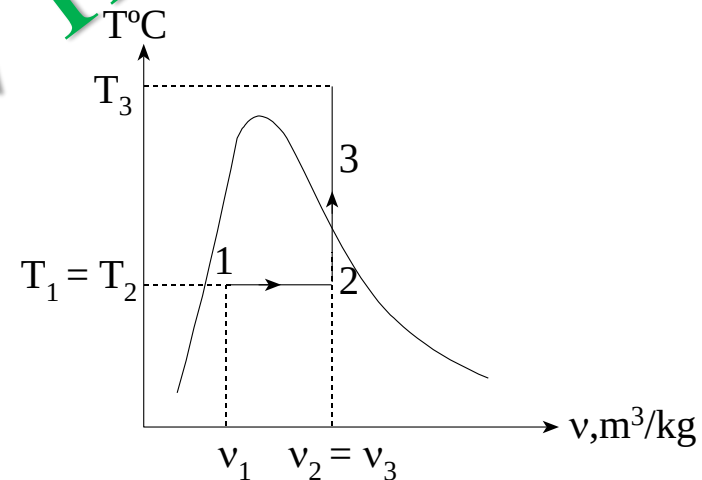
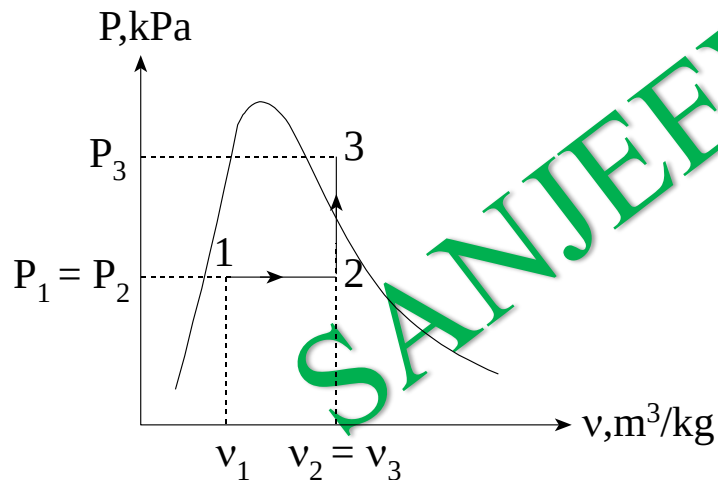
continued

Then applying linear interpolation for pressure,

$$P_3 - P_a = \frac{P_b - P_a}{(v_g)_b - (v_g)_a} [v_3 - (v_g)_a]$$

$$\therefore P_3 = P_a + \frac{P_b - P_a}{(v_g)_b - (v_g)_a} [v_3 - (v_g)_a]$$

$$= 600 + \frac{800 - 600}{0.3843 - 0.5137} (0.44142 - 0.5137) = 711.72 \text{ kPa}$$



continued

Now, total work transfer is given as

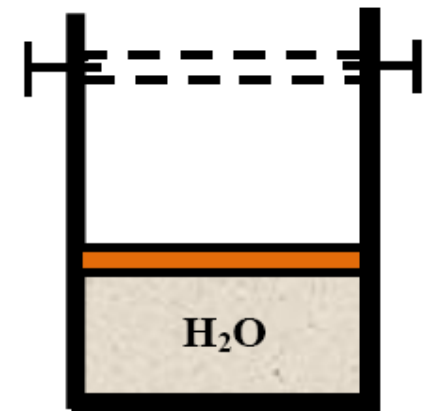
$$\begin{aligned} W &= W_{12} + W_{23} = P_2 (V_2 - V_1) + 0 \\ &= 125 (0.00062784 - 0.0003924) = 0.02943 \text{ kJ} \end{aligned}$$

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Sample Problem 3.4

A piston cylinder arrangement shown in figure below contains 0.2 kg of water initially at a pressure of 150 kPa with a quality of 40 %. The system is heated to a position where the piston is locked, and then cooled till it becomes a saturated vapor at a temperature of 60°C. Sketch the process on $P - v$ and $T - v$ diagrams and determine the total work transfer.

Solution:



continued

Given, mass of water (m) = 0.2 kg

Initial state: $P_1 = 150$ kPa, $x_1 = 40\% = 0.4$

Final state: $T_{\text{final}} = 60^\circ \text{C}$

Referring to Table A2.1, $T_{\text{sat}} (150 \text{ kPa}) = 111.38^\circ \text{C}$, $v_l (150 \text{ kPa}) = 0.001053 \text{ m}^3/\text{kg}$, $v_{lg} (150 \text{ kPa}) = 1.1584 \text{ m}^3/\text{kg}$, $v_g (150 \text{ kPa}) = 1.1595 \text{ m}^3/\text{kg}$,

$\therefore$ Specific volume at state 1 (v_1) = $v_l + x_1 v_{lg}$
 $= 0.001053 + 0.4 \times 1.1584 = 0.464413 \text{ m}^3/\text{kg}$

Process 1 – 2: Constant pressure heating process

State 2: $P_2 = 150$ kPa, superheated steam

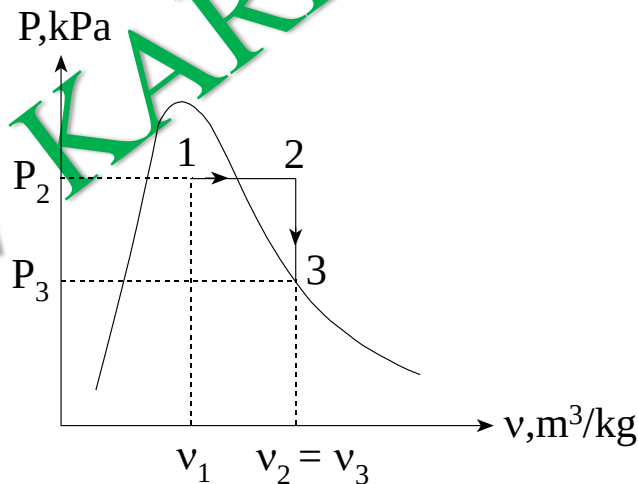
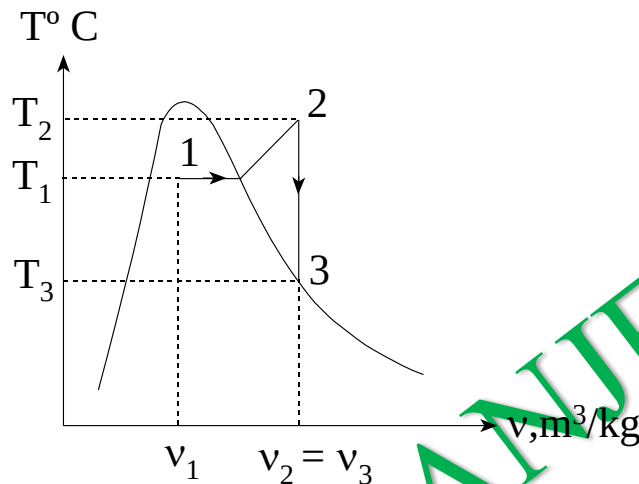
Process 2 – 3: Constant volume cooling process

State 3: $T_3 = 60^\circ \text{C}$, saturated vapor

continued

Referring to Table A2.2, $v_g (80^\circ\text{C}) = 7.6743 \text{ m}^3/\text{kg}$, $P_{\text{sat}} (60^\circ\text{C}) = 19.932 \text{ kPa}$

Hence, $v_3 = 7.6743 \text{ m}^3/\text{kg}$, $P_3 = 19.932 \text{ kPa}$



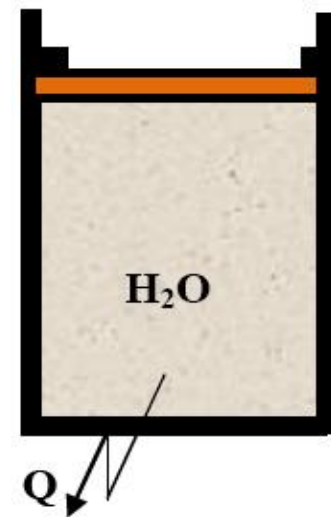
Then, total work transfer is given as

$$\begin{aligned} W &= W_{12} + W_{23} = 0 + P_2 (V_2 - V_1) = mP_2 (v_2 - v_1) \\ &= 0.2 \times 150 (7.6743 - 0.464413) = 216.297 \text{ kJ} \end{aligned}$$

Sample Problem 3.5

A piston cylinder arrangement shown in figure below contains 1 kg of water initially at a pressure of 1 MPa and a temperature of 500°C. The water is cooled until it is completely converted into the saturated liquid. It requires a pressure of 400 kPa to support the piston. Sketch the process on $P - v$ and $T - v$ diagrams and determine the total work transfer.

Solution:



continued

Given, mass of water (m) = 1 kg

Initial state: $P_1 = 1\text{MPa} = 1000\text{ kPa}$, $T_1 = 500^\circ\text{C}$

Pressure required to support the piston (P_{support}) = 400 kPa

Referring to Table A2.1, $T_{\text{sat}} (1000\text{ kPa}) = 179.92^\circ\text{C}$. Here, $T > T_{\text{sat}}$, hence it is a superheated vapor.

Referring to Table A2. 4, $v_g (500^\circ\text{C}) = 0.3541\text{ m}^3/\text{kg}$

$\therefore$ Specific volume at state 1 (v_1) = $0.3541\text{ m}^3/\text{kg}$

Process 1 – 2: Constant volume cooling process

State 2: $P_2 = 400\text{ kPa}$, $v_2 = 0.3541\text{ m}^3/\text{kg}$

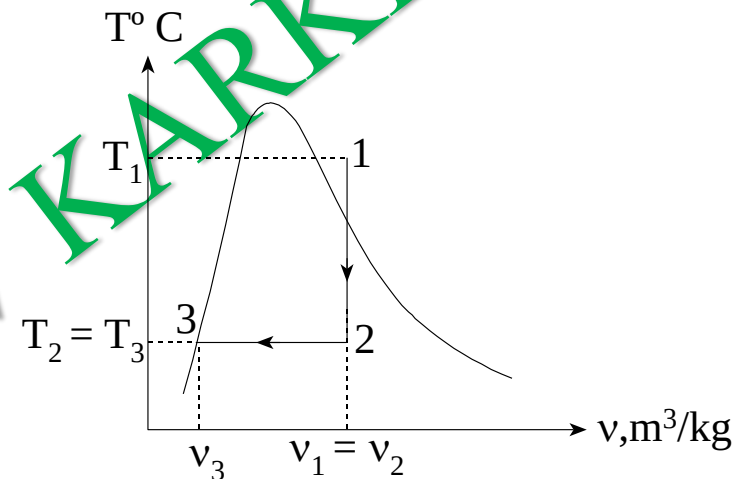
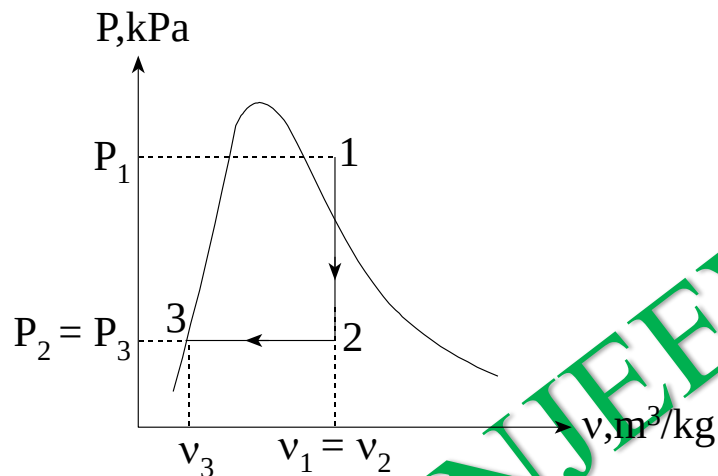
Referring to Table A2.1, $v_l (400\text{ kPa}) = 0.001084\text{ m}^3/\text{kg}$, $v_{lg} (400\text{ kPa}) = 0.4612\text{ m}^3/\text{kg}$, $v_g (400\text{ kPa}) = 0.4625\text{ m}^3/\text{kg}$.

Here, $v_l < v < v_g$, hence it is a two phase mixture.

continued

Process 2 – 3: Constant pressure cooling process

State 3: $P_3 = 400$ kPa, saturated liquid, $v_3 = 0.001084$ m³/kg



Hence, total work transfer is given as

$$\begin{aligned}
 W &= W_{12} + W_{23} = 0 + P_2 (V_3 - V_2) = mP_2 (v_3 - v_2) \\
 &= 1 \times 400 (0.001084 - 0.3541) = -141.21 \text{ kJ}
 \end{aligned}$$

Sample Problem 3.6

A piston cylinder arrangement shown in figure below contains 2 kg of water initially at a pressure of 200 kPa and a temperature of 50°C. Heat is added until the piston reaches the upper stops where the total volume is 1.5 m³. It takes a pressure of 600 kPa to lift the piston. Sketch the process on $P - v$ and $T - v$ diagrams and determine the final temperature and the work transfer.

(TU 2075 Baishakh)

Solution:



Given, Mass of water (m) = 2 kg

State 1: $P_1 = 200 \text{ kPa}$, $T_1 = 50^\circ \text{C}$

Final state: $V_{\text{final}} = 1.5 \text{ m}^3$

Pressure required to lift the piston, $P_{\text{lift}} = 600 \text{ kPa}$

Referring to Table A2.1, $T_{\text{sat}} (200 \text{ kPa}) = 120.24^\circ \text{C}$.

Here, $T_1 < T_{\text{sat}}$, hence it is a compressed liquid. Then, referring to Table A2.2 (since 200 kPa is not available in Table A2.3), $v_f (50^\circ \text{C}) = 0.001012 \text{ m}^3/\text{kg}$

$\therefore$ Specific volume at state 1 (v_1) = $0.001012 \text{ m}^3/\text{kg}$

And, specific volume at final state (v_{final}) = $\frac{V_{\text{final}}}{m} = \frac{1.5}{2} = 0.75 \text{ m}^3/\text{kg}$

Process 1 – 2: Constant volume heating process

State 2: $P_2 = 600 \text{ kPa}$, $v_2 = 0.001012 \text{ m}^3/\text{kg}$

continued

Referring to the Table A2.1, v_l (600 kPa) = 0.001101 m³/kg. Here $v_l > v$, hence it is a compressed liquid.

Process 2 – 3: Constant pressure heating process

State 3: $P_3 = 600\text{kPa}$, $V_3 = 1.5\text{ m}^3$, $v_3 = 0.75\text{ m}^3/\text{kg}$

Referring to Table A. 2.1, v_g (600kPa) = 0.31560 m³/kg.

Here, $v_3 > v_g$, hence it is a superheated vapor. Then, referring to Table A2.4, specific volumes of superheated steam which includes the specific volume 0.75 m³/kg and corresponding temperature are listed as

$T, ^\circ\text{C}$	$v_g, \text{m}^3/\text{kg}$	
700	0.7472	(a)
T_2	0.75	
750	0.7859	(b)

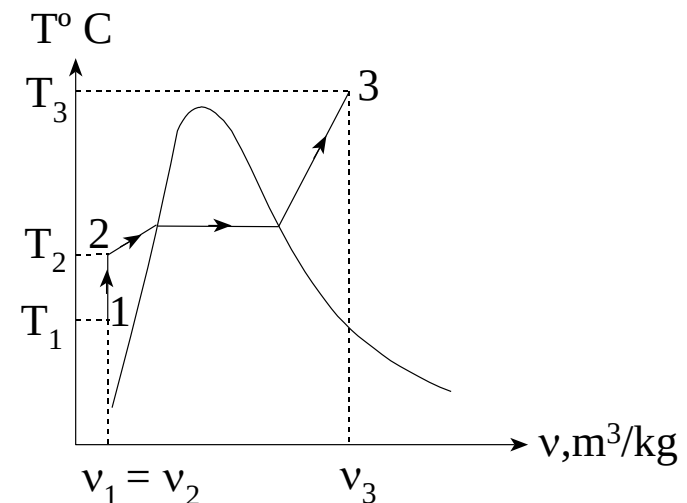
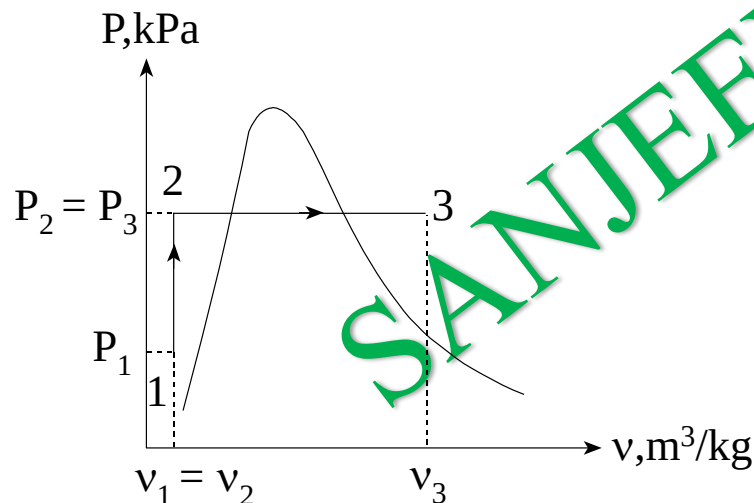
continued

Then applying linear interpolation for temperature,

$$T_3 - T_a = \frac{T_b - T_a}{(v_g)_b - (v_g)_a} [v_3 - (v_g)_a]$$

$$\therefore T_3 = T_a + \frac{T_b - T_a}{(v_g)_b - (v_g)_a} [v_3 - (v_g)_a]$$

$$= 700 + \frac{750 - 700}{0.7859 - 0.7472} (0.75 - 0.7472) = 703.65^\circ \text{C}$$



continued

Then, total work transfer is given as

$$\begin{aligned} W &= W_{12} + W_{23} = 0 + P_2 (V_3 - V_2) = mP_2 (v_3 - v_2) \\ &= 2 \times 600 \times (0.75 - 0.001012) = 898.786 \text{ kJ} \end{aligned}$$

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Sample Problem 3.7

A piston cylinder arrangement shown in figure below contains water initially at $P_1 = 100$ kPa, $x_1 = 0.8$ and $V_1 = 0.01$ m³. When the system is heated, it encounters a linear spring ($k = 100$ kN/m). At this state volume is 0.015 m³. The heating continues till its pressure is 200 kPa. If the diameter of the piston is 0.15 m, determine

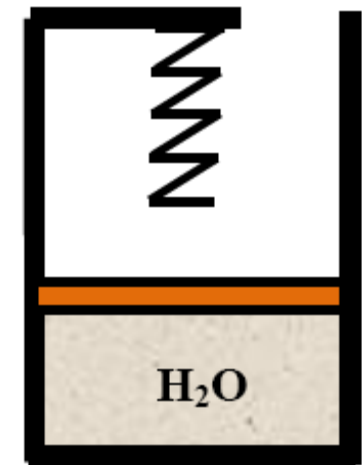
(a) the final temperature, and

(b) the total work transfer.

Also sketch the process on $P - v$ diagram.

(TU 2068 Magh, TU 2071 Chaitra)

Solution:



continued

Given,

State 1: $P_1 = 100 \text{ kPa}$, $x_1 = 0.8$, $V_1 = 0.01 \text{ m}^3$

State 2: $V_2 = 0.015 \text{ m}^3$

State 3: $P_3 = 200 \text{ kPa}$

Spring constant (k) = 100 kN/m

Diameter of the piston (d_p) = 0.15 m

$$\therefore \text{Area of the piston } (A_p) = \frac{\pi(d_p)^2}{4} = \frac{\pi}{4} \times 0.15^2 = 0.01767 \text{ m}^2$$

Referring to Table A2.1. $T_{\text{sat}} (100 \text{ kPa}) = 99.632^\circ\text{C}$, $v_l (100 \text{ kPa}) = 0.001043 \text{ m}^3/\text{kg}$, $v_{lg} (100 \text{ kPa}) = 1.6933 \text{ m}^3/\text{kg}$

Specific volume at state 1 (v_1) = $v_l + x_1 v_{lg}$
 $= 0.001043 + 0.8 \times 1.6933 = 1.35568 \text{ m}^3/\text{kg}$

continued

$$\therefore \text{Mass of H}_2\text{O (m)} = \frac{V_1}{v_1} = \frac{0.01}{1.35568} = 0.007376 \text{ kg}$$

Process 1 – 2: Constant pressure heating process

State 2: $P_2 = 100 \text{ kPa}$, $V_2 = 0.015 \text{ m}^3$

$$\therefore \text{Specific volume at state 2 (} v_2 \text{)} = \frac{V_2}{m} = \frac{0.015}{0.007376} = 2.0336 \text{ m}^3/\text{kg}$$

Here, $v_2 > v_g$, hence, it is superheated steam.

Process 2 – 3: Pressure and volume varies linearly

State 3: $P_3 = 200 \text{ kPa}$

But, the pressure due to spring only (P_{spring}) = $P_3 - P_2 = 200 - 100 = 100 \text{ kPa}$

Then, spring force is given by

$$F_{\text{spring}} = P_{\text{spring}} \times A_p = kx$$

continued

$$\therefore \text{Compression of spring (x)} = \frac{P_{\text{Spring}} \times A_p}{k} = \frac{100 \times 0.01767}{100} = 0.01767 \text{ m}$$

Thus, volume at state 3 is given as

$$V_3 = V_2 + A_p x = 0.015 + 0.01767 \times 0.01767 = 0.015312 \text{ m}^3$$

And, specific volume at state 3 is given as

$$v_3 = \frac{V_3}{m} = \frac{0.015312}{0.007376} = 2.076 \text{ m}^3/\text{kg}$$

Referring to Table A2.1, $v_g (200 \text{ kPa}) = 0.08859 \text{ m}^3/\text{kg}$. Here, $v_3 > v_g$, hence it is a superheated vapor.

Then, referring to Table A2.4, specific volume of superheated vapor which includes the specific volume $2.075 \text{ m}^3/\text{kg}$ and corresponding temperatures are listed as

continued

$T, ^\circ\text{C}$	$v_g, \text{m}^3/\text{kg}$	
600	2.0130	(a)
T_3	2.075	
650	2.1287	(b)

Then applying linear interpolation for temperature,

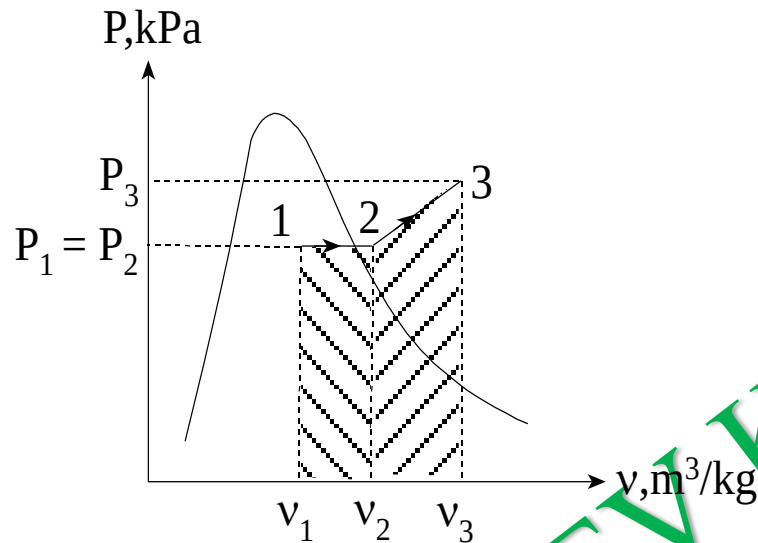
$$T_3 - T_a = \frac{T_b - T_a}{(v_g)_b - (v_g)_a} [v_3 - (v_g)_a]$$

$$\therefore T_3 = T_a + \frac{T_b - T_a}{(v_g)_b - (v_g)_a} [v_3 - (v_g)_a]$$

$$= 600 + \frac{650 - 600}{2.1287 - 2.0130} (2.076 - 2.0130) = 627.226^\circ\text{C}$$

The final temperature, $T_3 = 627.226^\circ\text{C}$

continued

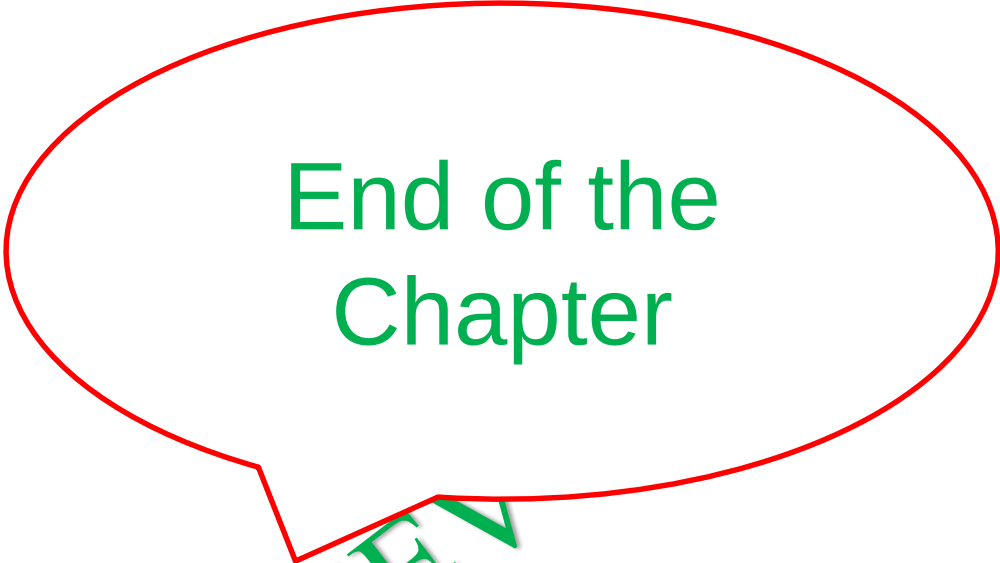


Total work transfer is given as

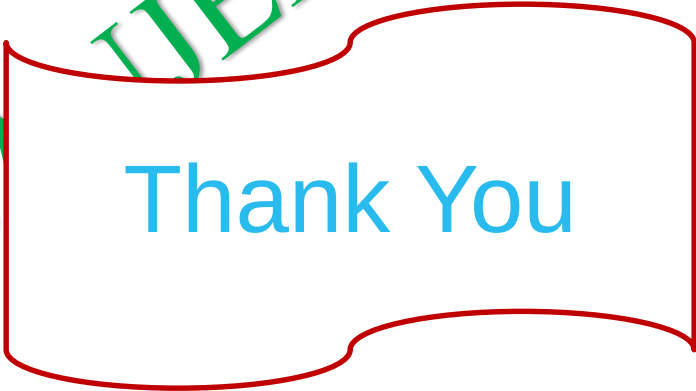
$$\begin{aligned}
 W &= W_{12} + W_{23} = P_1 (V_2 - V_1) + \frac{1}{2} (P_2 + P_3) (V_3 - V_2) \\
 &= 100 (0.015 - 0.01) + \frac{1}{2} (100 + 200) (0.015312 - 0.015) = 0.5468 \text{ kJ}
 \end{aligned}$$

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End of the
Chapter



Thank You